Tangible Worlds:
Physical Arrangement as Spatial Prompting for Generative 3D Environments

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ABSTRACT

This dissertation asks what happens when the spatial structure of a generated 3D scene is arranged by hand rather than described in a prompt. Tangible Worlds is a tabletop system in which ordinary physical props act as persistent handles for virtual objects. Their position and proximity come from the physical arrangement, while language assigns what they mean. A generative image-to-3D pipeline supplies their visual form.

The project was developed through practice-based research and iterative making. Successive prototypes were tested on the working hardware, with measurements, logged sessions and controlled comparisons used to examine failures and guide design changes. This process showed that separating physical structure from semantic meaning is technically achievable, but that the boundary between human and computational authorship cannot be treated as a single division. Tracking, prompt shaping, scale control and speech recognition each exposed cases where an automatic process could silently take over a decision that the interaction presented as belonging to the person. This led to a property-by-property account of authorship: computational delegation can remain compatible with authored control when its boundary is explicit.

The final system allows position, meaning, scale and atmosphere to be revised independently, although generated appearance is less flexible because assets persist once generated. No creator study was completed, so the dissertation does not claim that the system increased creativity, agency or ownership. Creative exploration is instead treated as an affordance of the architecture. The contribution is a working artefact and a documented design account of how physical arrangement, linguistic intent and generative processes can share responsibility for constructing a 3D environment.

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1 Introduction

Generative 3D systems increasingly allow scenes to be created from text or images. TabletopGen, for example, generates instance-level 3D tabletop scenes from either a textual description or a single image, while SpaceBlender uses images of users' physical surroundings to construct shared virtual environments (Numan *et al.*, 2024; Wang *et al.*, 2026). In these examples, spatial composition is derived from textual or visual input rather than directly manipulated as a physical arrangement. For a task concerned with where things are placed and how they relate, layout therefore becomes something represented to the system rather than something arranged by hand.

Tangible interfaces offer a different starting point. Systems such as *Urp* coupled physical architectural models to computational simulations on a tabletop, and Kim and Maher's comparison of tangible blocks with a desktop interface found that designers in the tangible condition attended more to spatial relationships between elements (Underkoffler and Ishii, 1999; Kim and Maher, 2008). More recent work has begun to combine tangible interaction with generative models. Tangible Diffusion, for example, lets students arrange prepared physical elements that condition AI-generated 2D artwork (Han, Tang and Wang, 2024). The gap addressed here is therefore not tangible interaction or generative 3D on their own, but a particular division of responsibility between them: physical arrangement supplies the spatial structure of a generated 3D scene, while language separately assigns open semantic meaning to the things being arranged.

Tangible Worlds explores that division. Ordinary physical props are tracked on a tabletop as persistent handles. Their position and proximity come from the physical arrangement, while their semantic identity is assigned through language. A prop does not need to resemble what it represents: the same physical object can stand for a house, a horse or a mountain. Once assigned, that meaning remains attached while the prop moves, and the generated asset follows its position. In this dissertation I use **spatial authorship** to mean authorship exercised through arranging and revising the spatial structure of a generated environment, while semantic intent is supplied separately through language.

The research question remained unchanged throughout the project:

How might tangible interfaces that translate physical arrangements into generative 3D environments support new forms of spatial authorship and creative exploration?

The word *support* is treated cautiously here. No participant study was ultimately conducted, so this dissertation does not claim that creators found the system intuitive, felt greater ownership, explored more freely or preferred one input method. Instead, it examines what forms of spatial control and exploratory action the built mapping makes available, where that mapping fails, and which decisions remain with the person or shift to the generative system. The contribution is therefore a working artefact and a documented design account of separating spatial structure, semantic intent and generated appearance, including the cases where that separation could not be maintained cleanly.

2 Background and related work

2.1 Tangible and miniature control of space

The idea that a smaller representation can stand in for a larger digital space predates current generative systems. Stoakley, Conway and Pausch's Worlds in Miniature (WIM) gives a user a hand-held miniature copy of an immersive virtual environment. Manipulating an object in the miniature changes its corresponding life-size virtual object (Stoakley, Conway and Pausch, 1995). WIM is not a physical tabletop system, but it establishes a useful principle for this project: spatial structure can be manipulated through a proxy rather than described indirectly.

Urp moves that principle onto a physical workbench. Architectural models placed on a table are coupled to projected simulations of shadows, reflections and wind, so moving or rotating a model changes the corresponding simulation (Underkoffler and Ishii, 1999). The architectural models relevant here already carry stable meanings: a building model represents a building. The reacTable also makes spatial configuration computationally consequential. Each puck has a dedicated sound-generation, modification or control function, while connections depend on object type, affinity and proximity (Jordà *et al.*, 2007). For these objects, the mapping between tangible and digital function is defined by the system rather than assigned freely in language during use.

Macaranas, Antle and Riecke distinguish physical attributes such as size, weight and texture from spatial properties such as proximity and orientation when designing tangible mappings (Macaranas, Antle and Riecke, 2012). Tangible Worlds makes a deliberate choice inside that design space: position and proximity remain physical, while semantic identity is not inferred from a prop's appearance, size or category.

There is empirical reason to take that spatial role seriously, but the evidence is narrow. Kim and Maher compared a tabletop interface using 3D blocks with a desktop interface for architectural design. Their final protocol analysis covered seven designers and found greater attention to spatial relationships in the tangible condition, alongside more relocation of objects while testing configurations (Kim and Maher, 2008). They explicitly caution against generalising the result to all designers. Their task was also deterministic and used the same set of 30 furniture models in both conditions, so it cannot establish how creators would respond to generative output. It instead supports the narrower premise that tangible manipulation can change how spatial relations enter a design process.

2.2 Tangible input and generative output

Tangible Diffusion is a close precedent because physical construction directly conditions generative output. Students arrange tangible building blocks, photograph the result, and use that image through ControlNet alongside prompts and models prepared by teachers to generate digital artwork (Han, Tang and Wang, 2024). The arrangement therefore contributes composition to the generated image rather than acting only as a controller.

Its division of responsibility is different from Tangible Worlds. Tangible Diffusion's teachers choose or design the tangible elements, prepare prompts and select or train the relevant models; in some experiments, physical shapes are deliberately designed to represent imagery such as mountains, trees and water. Tangible Worlds instead treats the prop as a semantically neutral handle whose meaning is assigned in language, then generates a 3D asset that remains attached to that handle as it moves. The difference is not tangible interaction

combined with AI, which Tangible Diffusion already demonstrates, but where semantic and spatial decisions are placed.

Recent systems narrow the 3D side of the gap further. Thing2Reality converts content from video streams and shared screens into manipulable 3D objects for XR collaboration (Hu *et al.*, 2025). SpaceBlender uses user-provided images of physical surroundings, depth estimation, mesh alignment and generative completion to blend those surroundings into shared virtual environments (Numan *et al.*, 2024). These systems connect physical context to generated or reconstructed 3D content, but do not use a live arrangement of semantically neutral tabletop props as the scene's spatial control structure.

2.3 Generative 3D scenes and the position of Tangible Worlds

TabletopGen makes the distinction sharper. It generates instance-level 3D tabletop scenes from text or a reference image, with individual objects made physically interactive within simulation. Its pipeline segments the reference, reconstructs individual objects, estimates their pose and scale, and assembles them into a collision-free scene (Wang *et al.*, 2026). Spatial layout is therefore recovered or generated from textual or visual input rather than continuously supplied by movable physical proxies.

Across the literature reviewed here, the component ideas around Tangible Worlds are established separately: miniature representations can act as spatial proxies; physical objects can control digital relationships; tangible arrangements can condition generative images; and text or images can produce 3D scenes. Among the closest precedents reviewed here, I did not identify a verified system that gives these responsibilities the same separation used in Tangible Worlds: physical arrangement supplies continuously revisable scene structure, independent natural-language input assigns open semantic meaning to physically neutral handles, and generative models supply their visual form.

This is narrower than claiming a new class of tangible or generative interface. It also produces a different design problem. In Urp and the reacTable, much of the mapping relevant to the interaction is known before use. In Tangible Worlds, the physical layout is explicit but semantic interpretation and generated appearance pass through probabilistic models. The question becomes how spatial control remains legible when spatial structure, semantic intent and generated appearance are handled by different parts of the system.

Human-computer co-creativity provides a useful comparison for this division of responsibility. Kantosalo and Toivonen (2016) distinguish alternating co-creativity from task-divided co-creativity, in which human and computational partners take specific roles within a creative process. They further describe concept definition, generation and evaluation as separable roles, while noting that real systems rarely fit rigid categories. Tangible Worlds is not presented here as a fully co-creative partnership, but the task-divided framing helps describe its separation of contributions: physical action determines spatial structure, language supplies semantic intent, and generative models determine much of the resulting visual form.

3 Methodology

Candy and Edmonds (2018) distinguish practice-based research, where a creative artefact forms part of the contribution to knowledge, from practice-led research, where the contribution concerns new understandings of practice. This project contains both. The artefact is central, but the contribution also includes what building and interrogating it established: meaning needed to be decoupled from the physical handle, height was too unstable to carry identity, a fixed global prompt did not generalise across the meanings tested, and explicitly specified scale proved safer than inferred scale. These findings came from making the system rather than existing independently of it.

This also aligns with Frayling's (1993) category of research through art and design, in which practical experimentation is documented and contextualised so that its results can be communicated. The method inside that frame was reflective practice through iterative making.

For six weeks the working loop was simple: build a change, run it on real hardware, record what happened, interpret the result, then change the design and test again. Schön (1983) describes reflection-in-action as beginning when practice produces a surprise that forces attention back onto the assumptions behind an action. The identity classifier provides a clear example. I initially tuned height thresholds by hand, then measured a stationary prop drifting from 20 to 28 mm and a handled prop reading 16, 26 and 40 mm across three placements. The problem stopped being the threshold and became the feature, so height was removed from the identity decision.

That observation, interpretation and design-change sequence is the standard used for Section 5. Implementation history is included only where it establishes something about the design problem.

Four kinds of evidence support the claims in this report. Source inspection establishes what the implemented system does. Executed measurement covers feature captures, classifier analyses, scale tests and speech-recognition studies. Live hardware sessions with contemporaneous captures support claims involving physical props, hands and occlusion. Controlled comparisons are used where an input could be held constant while one relevant component changed, such as reconstructing the same tree source image with SF3D and SPAR3D.

Three rules govern how that evidence becomes a claim. A preference is stated only where a comparison supports it rather than from impression alone. A single observation remains a single observation rather than becoming a reliability claim. Where supporting evidence could not be located, the claim was withdrawn. Several statements from earlier project notes were removed during the final verification pass for that reason.

The evidential record consists of LOGS.docx, a dated record of 83 captioned development images, and PORTFOLIO.zip, containing 1,020 screenshots, masks, comparison grids and terminal captures indexed by section and content. Code is retained separately in the project repository. Figures in Section 5 are selected as evidence for specific claims rather than as decoration. This use of an indexed, annotated collection follows the evidential logic of Gaver and Bowers (2012): the selected artefact can be read together with the surrounding material rather than in isolation.

Rejected prototypes are also treated as evidence where they narrowed the design space. Lim, Stolterman and Tenenberg (2008) describe prototypes as filters through a design space rather

than incomplete versions that matter only when they succeed. The rejected reconstruction models, terrain approaches and automatic scale mechanisms therefore remain relevant where their failures changed the subsequent design.

This evidence has a hard boundary. Build evidence can establish that a mapping was implemented, that a sensing feature was too unstable for a particular role, that generation introduced variability, that movement and meaning were decoupled, and that one division between physical structure and linguistic semantics is technically achievable. It cannot establish that creators found the mapping intuitive, experienced greater ownership, explored more freely or preferred one input modality. Where Section 6 discusses authorship, it describes how the architecture distributes the means of authorship, not how anyone experienced that distribution.

There is no controlled participant study. A within-subjects comparison of typed and spoken input was designed and instrumented but not run. By the end of development, stochastic object generation meant that each condition would include different generated outputs, making a clean relative comparison difficult to interpret. This did not make user inquiry unnecessary: qualitative observation or think-aloud sessions would still have been valuable, but were not completed within the project schedule. As discussed in Section 2.1, Kim and Maher's deterministic tabletop comparison motivates studying spatial manipulation but does not supply evidence about this generative system. The unused study instrumentation remains in the interface, which is why its URL accepts a condition and participant identifier.

Not every part of the system is probabilistic. Five atmosphere commands were each run three times from the same starting state and their resulting frames compared pixel by pixel. Section 5.6 reports that deterministic part of the evidence separately.

4 System overview

Tangible Worlds separates spatial structure from semantic meaning. Physical props determine where objects are and how they relate spatially; language determines what those props mean and controls the surrounding environment. The two streams meet at the server that maintains scene state (Figure 4.1).

An overhead iPhone streams LiDAR depth to the MacBook. After table calibration, connected components above the table plane become tracked objects with persistent numeric ids. The sensing layer records geometry but assigns no semantic labels. A mapping layer converts those objects into normalised 2D positions and relations such as distance, direction, proximity and clustering.

Meaning is assigned separately through typed or spoken language. A local Qwen3-8B (Yang *et al.*, 2025) shapes an assigned meaning into an object prompt; SDXL Turbo (Sauer *et al.*, 2023) generates a source image and SPAR3D (Huang *et al.*, 2025) reconstructs it as a GLB mesh. Generation runs on a rented RTX 4090, while scene state and interaction remain on the laptop. Generated meshes are cached by meaning, so moving a prop changes its position without regenerating its appearance.

The same language model handles atmosphere through eleven numeric scene parameters plus a display-only mood label, which the renderer maps deterministically to lighting, sky, mist and weather. In voice mode, speech is transcribed locally with Parakeet TDT 0.6B v3 (NVIDIA, 2025) and read back through Pocket TTS (Rouard *et al.*, 2026) for confirmation before any

scene change is applied.

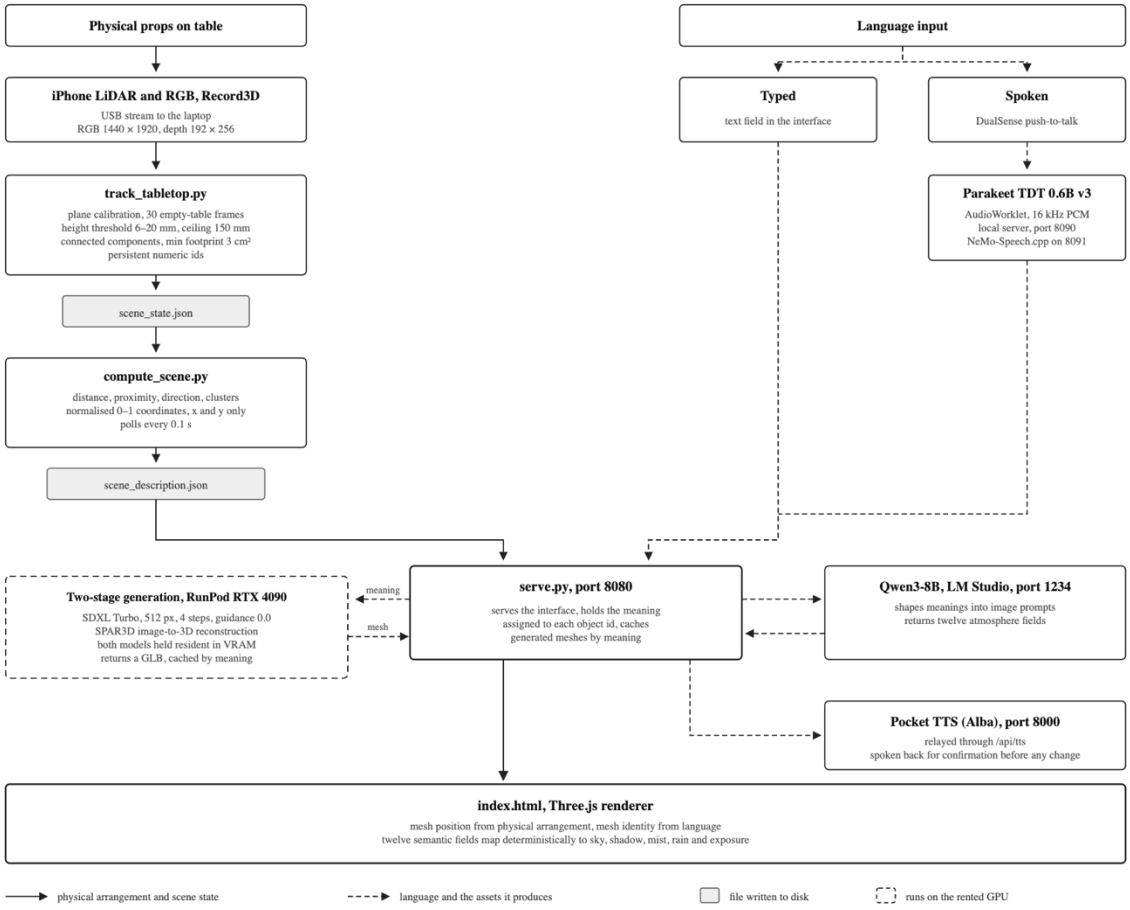


Figure 4.1 System architecture. Physical props are sensed as LiDAR depth, tracked as persistent ids and mapped into 2D spatial relations. Typed or spoken language assigns meaning and controls atmosphere; Qwen3-8B shapes commands, while SDXL Turbo and SPAR3D generate object meshes on a rented GPU. Both streams meet at the local scene server.

5 Development strands

The system developed iteratively across several overlapping strands rather than as a single sequence. The development is therefore grouped by subsystem. Each strand follows the same structure: the design problem, implementation, test evidence, and the decision that followed.

5.1 Tracking and object identity

5.1.1 From semantic recognition to geometric tracking

The sensing layer ultimately needed to answer three questions: whether something was on the table, where it was, and whether it was the same physical thing after movement. Semantic meaning belonged elsewhere.

The first prototype violated that separation. YOLO detected objects from the iPhone RGB stream and passed both ids and class labels into the browser scene.

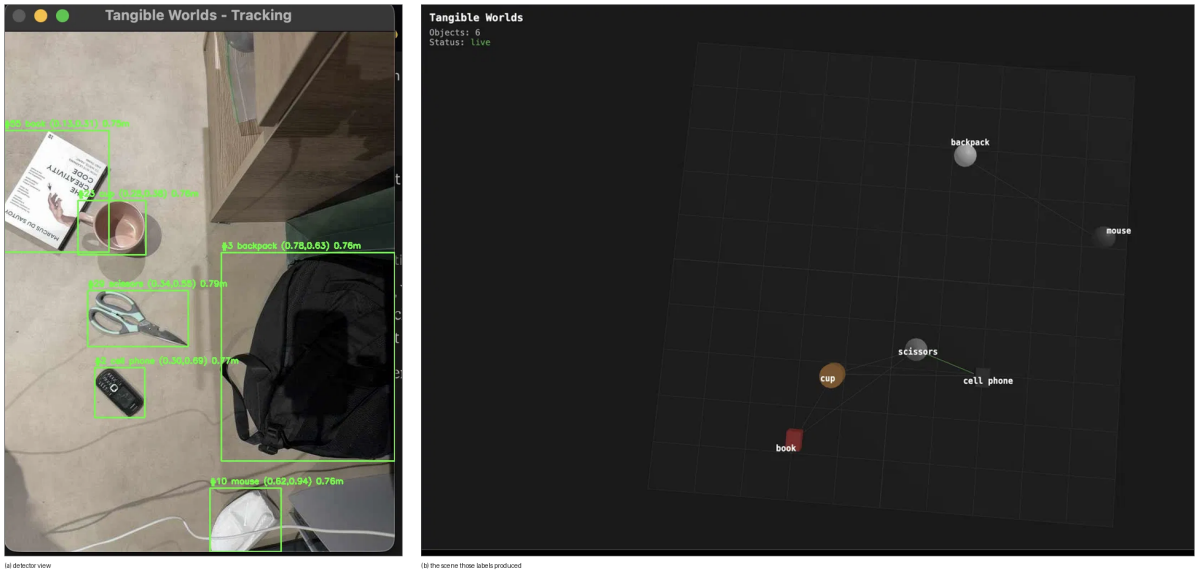


Figure 5.1 First working tracking loop. YOLO detections on the iPhone stream feed the browser scene directly. A face mask is classified as mouse and enters the generated scene under that label.

The misclassification was visible, but the larger problem was structural. The scene vocabulary was backpack, mouse, scissors, cup, cell phone and book because those were categories available to the detector, not meanings assigned through the interface. Improving recognition accuracy would not fix that. A system in which a person assigns meaning to a prop cannot let a detector decide the vocabulary in advance.

A later four-prop test confirmed the practical limitation from the other direction: at confidence 0.20 YOLO recognised only one of the tabletop props, while one prop, a watch, was outside COCO entirely (Lin *et al.*, 2014). The production tracker therefore removed object detection and moved semantic assignment into language.

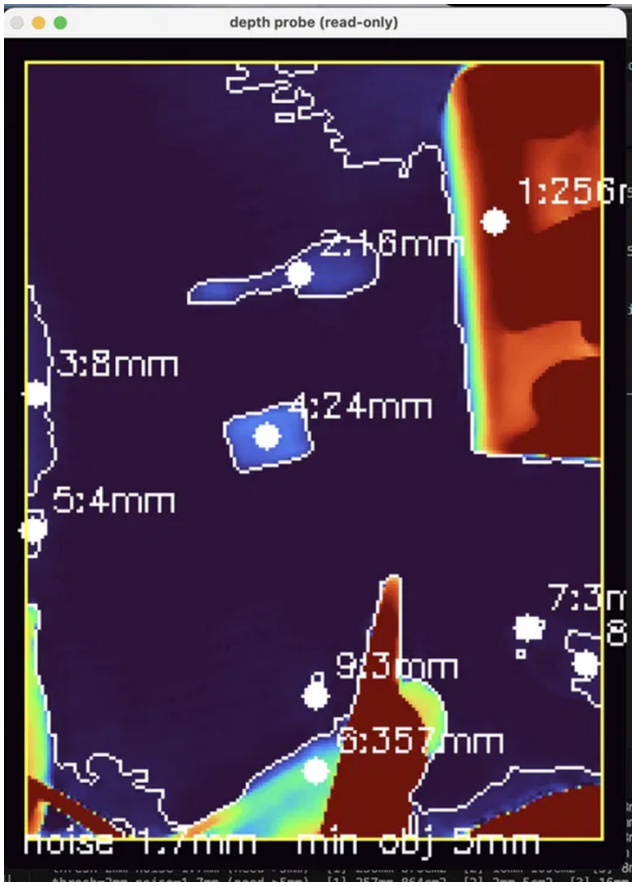


Figure 5.2 Depth probe during the move to geometry-only detection. Connected components above the fitted table plane include both real foreground and false regions from the wall, table edge, surface variation and sensor noise, motivating ROI, height and footprint constraints.

Depth became responsible only for geometry. The system fits a plane to an empty table and treats connected regions above it as candidate objects.

Calibration uses 30 empty-table frames to derive a foreground threshold from measured residual noise. The threshold is clipped to 6–20 mm; calibration is rejected above 2.5 mm noise; components above 150 mm or below 3 cm² are discarded. These constraints decide what counts as geometry without assigning what any prop means.

5.1.2 Identity features and classifier design

Removing semantic recognition left a harder problem: keeping the same numeric identity attached to a prop as it moved.

Height initially looked useful because it came directly from the depth component. Hardware measurements rejected that assumption.

What was compared	Footprint area	Reported height
One prop sitting still on the table, not touched, across a single session	stable	drifted from 20 mm to 28 mm
One prop picked up and put back down, measured three times	not recorded	16 mm, then 26 mm, then 40 mm
One prop measured before a move and again after it	agreed within 1%	changed substantially

Table 5.1 Measurements that led to height being removed from identity matching. Reported height changed substantially both while stationary and after handling, while footprint was substantially more stable in the recorded comparisons.

A stationary prop drifted from 20 to 28 mm, and another read 16, 26 and 40 mm after repeated handling. Height therefore remained useful for geometric rejection but was removed from identity matching. Colour alone also failed: rotating one prop onto another face changed its RGB histogram enough to reject it even though its footprint remained within 7%.

Feature combinations were therefore measured rather than continuing to tune single-feature thresholds. A hardware capture produced 986 labelled rows across four props, with 24 measurements per frame and no hands present. The final feature vector uses elongation, circularity, the first two Hu moments and five colour statistics. Hu moments contribute scale- and rotation-invariant outline information in the image (Hu, 1962). Features are rescaled before a nearest-centroid comparison so their units do not determine their influence.

The classifier remains deliberately lightweight. A small set of prop-specific examples defines the centroids, consistent with the interactive-machine-learning framing described by Fiebrink and Cook (2010), rather than using a pretrained recognition vocabulary.

Two state rules protect that learned set. An unmatched blob remains tentative until thirty consecutive detections, following the general tracking-by-detection idea of delaying public identity until a track is established (Bewley *et al.*, 2016). Learning is then explicitly locked after the placed props have been sampled, preventing later hand fragments or unexpected blobs from becoming new permanent identities.

5.1.3 Refusal thresholds and identity uncertainty

The remaining question was where matching should stop and the tracker should admit uncertainty.

Table 5.2 re-analyses the archived four-prop capture. Three props provide centroids from the first 135 frames of their sequences; their 330 later frames measure recognition, while 251 frames from the fourth prop measure refusal.

Refusal threshold	Later frames of the three learned props, correctly identified	Frames of the fourth prop, correctly refused
0.50	275 of 330	251 of 251
0.55, shipped	286 of 330	236 of 251
0.60	291 of 330	189 of 251
0.80	328 of 330	37 of 251

Table 5.2 Retrospective threshold analysis from the archived four-prop capture. Centroids use the first 135 frames of three props; later frames measure recognition and the fourth prop measures refusal. Feature normalisation used the full capture and the fourth prop informed the historical threshold choice, so this is parameter re-analysis rather than independent validation.

That qualification matters. The feature normalisation was fitted across all 986 rows, including evaluation frames, and the fourth prop had originally contributed to choosing the threshold. The numbers therefore explain the setting; they do not establish generalisation to an unseen prop.

The important design choice is the asymmetry between errors. Refusal is visible and recoverable: the object remains unidentified. Silent identity transfer is harder to notice because an existing meaning can move onto the wrong physical handle.

The submitted tracker uses a threshold of 0.55. Re-analysis showed that this does not fully follow the stated preference for visible refusal. Moving from 0.50 to 0.55 gains 11 correctly identified learned-prop frames but loses 15 correct refusals of the unfamiliar prop. If silent transfer is considered the more consequential error, 0.50 follows that principle better.

I therefore retain 0.55 as the documented shipped parameter rather than rewriting the history around the later analysis. The roster lock prevents new identities being created after learning, but it cannot guarantee that a blob will never be assigned to the wrong existing identity. Likewise, the recorded hardware session showed that no new identity appeared during 1,039 seconds of operation, but no unfamiliar prop was deliberately introduced, so it cannot establish refusal behaviour beyond the archived analysis.

The disagreement between the stated principle and the shipped parameter is itself a finding: a silent-risk setting can survive development precisely because it may produce no obvious visible symptom.

5.1.4 Occlusion, merges and identity recovery

Hands create a different tracking problem because they temporarily become some of the largest foreground components in the workspace.

A component touching the frame edge is treated as body geometry, and ordinary identity processing resumes only after five consecutive clear frames. While a hand is within 3 cm of a known prop, that prop remains latched to its last clean measurement rather than being

relearned during handling. This treats handling as contextual information rather than only as measurement noise, consistent with interactive perception (Bohg *et al.*, 2017) and with FlowPut's use of occlusion-resilient tracking for tangible objects (Riemann *et al.*, 2018).

Two props pushed together create the opposite problem: the depth image sees one connected component where two identities previously existed. Rather than creating a new identity, the tracker tests whether the merged component's area is consistent with the sum of two known footprints and keeps both identities alive through the merge. This is a smaller tabletop-specific response to the merged-measurement problem described by Henriques et al. (2011).

In the same hardware session, props with locked footprints of 83.3 and 101.0 cm² produced merged measurements of 184.6 and 183.2 cm² on two occasions, close to their 184.3 cm² combined area. The implementation uses a wider 15% tolerance because missing a genuine merge risks losing both identities, while a loose merge primarily delays separation.

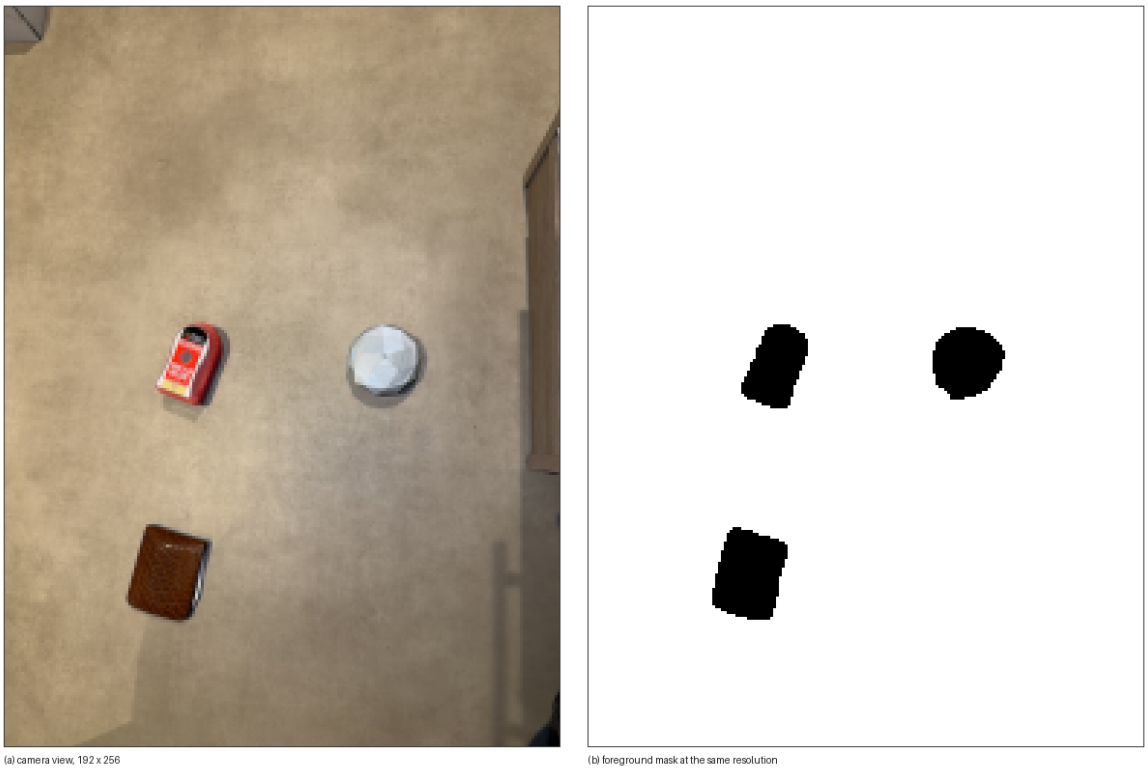


Figure 5.3 Recovery after a merge at 819 seconds in the recorded hardware session. The classifier could not confirm two re-separated components against the 0.55 threshold, scoring 0.76 and 0.77, so identity recovery fell back to position rather than appearance.

This episode exposes the classifier at its limit rather than only demonstrating a clean success. When appearance was insufficient, the system logged the uncertainty and used the remaining spatial evidence.

5.1.5 Final tracker configuration and limitations

Low-profile props remain the clearest unresolved sensing limitation. Four alternatives were tested: local depth background subtraction, static RGB background subtraction, naive RGB-depth fusion and a larger morphological close. None was adopted. The larger close still fragmented shallow objects and increased the risk of joining separate props.

The experiments also corrected my own initial interpretation. I first treated the limitation as small object size, but a stamp cover of roughly 7 cm² tracked successfully. The more defensible boundary is profile: shallow props can fluctuate around the table-plane threshold depending on material and placement.

The tracker was frozen once further sensing changes began to destabilise identity behaviour already supporting the rest of the system. In the subsequent hardware session it ran for 1,039 seconds and 26,503 frames. Three props were learned and the roster locked; sixteen hand episodes and two merges occurred without a new identity being created. Six uncertain frames were deliberately saved, including four post-lock cases that the tracker declined to name.

What this strand established for the rest of the project is therefore not perfect tracking. It established a sensing layer whose public output is numbered physical geometry rather than semantic categories, with explicit mechanisms for uncertainty and known limits around identity recovery and low-profile objects. That separation makes it possible for a meaning assigned later in language to remain attached to a movable physical handle.

5.2 Semantic assignment and the open vocabulary

5.2.1 Decoupling physical props from semantic meaning

A reconstruction model shown the physical prop would reconstruct the prop itself. If a mug on the table could only become a virtual mug, the tabletop would function as a scanner rather than as a spatial interface. Meaning therefore had to be supplied independently of physical appearance.

The assignment layer was built before it was connected to real generation. A tracked object can receive a meaning, enter a temporary generating state and then become ready. Moving it changes only its position; the assigned meaning remains attached and no regeneration occurs. Position and semantics therefore meet in the same object record while remaining controlled by different channels.

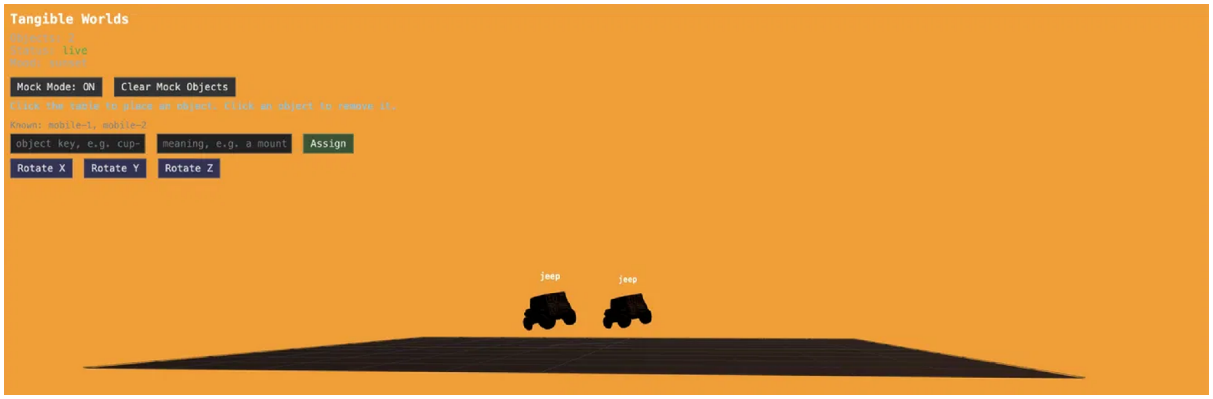


Figure 5.4 Two object identities assigned the same meaning, jeep, in mock mode. Both retain separate positions while referencing the same cached jeep.glb asset, demonstrating that semantic identity is independent of physical object identity.

The cache is keyed by meaning rather than by tracked object, so separate handles can share one generated asset while remaining independently movable. Nothing about the prop's photograph, colour or shape enters the semantic assignment. By that stage of the pipeline, the physical object is an id and a position; its meaning comes from language.

5.2.2 Open vocabulary and fixed-prompt limitations

Open semantic assignment immediately created a difficult trade-off. Early tests found that bounded concepts such as mountain and jeep could reconstruct coherently while dense forest failed. My first response was to consider restricting the permitted vocabulary to meanings the pipeline handled well.

I reversed that decision. A whitelist would recreate the problem exposed by YOLO in Section 5.1: the system would decide in advance what the person was allowed to mean. The source of the restriction would have changed from COCO to my own list, but the authorship problem would remain. I therefore kept the vocabulary open and treated unsupported meanings as visible limitations rather than excluding them silently.

A second restriction was less obvious because it existed inside the image prompt. Every meaning was initially followed by the same framing suffix:

```
low poly 3d model of <meaning>, single centered object, solid white background, isometric, game asset, blender render, sharp flat faces
```

That worked for some meanings and failed on others.



(a) 20:14 "mountain" (b) 20:24 "tree"

Figure 5.5 The same fixed suffix applied to mountain and tree ten minutes apart. Mountain produced a bounded single object; tree produced four cropped instances. No seed was fixed, so the figure records an observed failure rather than a controlled comparison.

I initially suspected terms such as game asset and isometric of encouraging reference-sheet layouts, but I did not isolate that mechanism. The defensible finding is narrower: one global template did not generalise across the ordinary meanings I tried. The problem was therefore not only keeping the vocabulary open, but trying to frame unknown future meanings with one prompt written in advance.

5.2.3 Adaptive prompt shaping

CANONICAL_VIEW_SUFFIX was removed and replaced with shape_prompt, which asks the local Qwen3-8B to construct a framing prompt separately for each assigned meaning. The model is allowed to decide presentation but not semantic content.

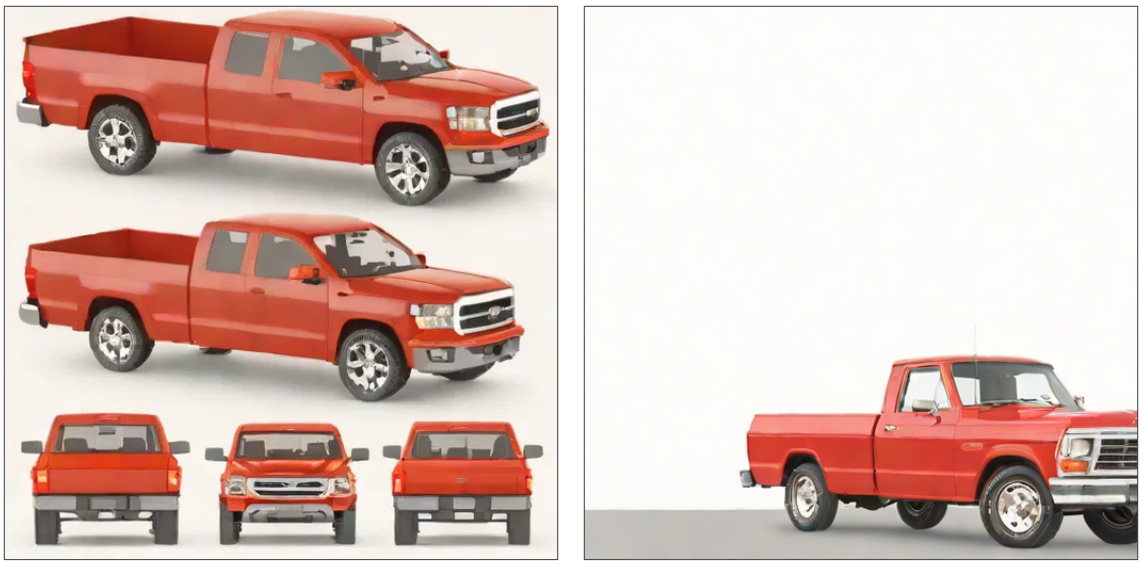
The key constraints were:

Keep the supplied subject unchanged; do not add species, materials, colours or semantic detail. Describe one fully visible centred subject from a three-quarter view, with clear surrounding margin and a plain studio background. Describe one photograph rather than a sheet or multiple views.

The new approach adapts framing per meaning while keeping the subject fixed. A small probe compared Llama 3.2 3B (Meta, 2024) with Qwen3-8B (Yang *et al.*, 2025) on four meanings. Llama leaked subject colour into the background and changed cliff into cliff wall; Qwen followed both tested constraints on all four. In this probe, Qwen better preserved the intended semantic boundary and was therefore retained for prompt shaping. Four meanings are too few to support a general comparison between the models.

That distinction matters because semantic elaboration would move authorship across the boundary established in 5.2.1. If the assigned meaning is cliff, the shaping model may decide framing and background, but it should not change the subject to cliff wall.

Two later vehicle failures produced further framing rules.



(a) five views instead of one photograph

(b) subject cut off at the frame edge

Figure 5.6 Two Stage 1 vehicle failures that informed prompt shaping. One pickup appears as a five-view reference sheet; another is cropped at the frame edge. These observations motivated the one-exposure and fully-visible-with-margin constraints.

The tests did not establish why the image model produced either failure. They established what the downstream reconstructor required: one complete subject, presented once, with a usable boundary.

5.2.4 Representational limits and caching

Keeping semantic vocabulary open does not mean every meaning is representable by an object-generation pipeline.



(a) generated image (b) the mesh reconstructed from it

Figure 5.7 Dense forest. The Stage 1 image has no bounded single subject, and reconstruction produces a warped sheet. The failure illustrates the mismatch between an environment-scale meaning and a pipeline designed to generate discrete objects.

Across the meanings tested, bounded single subjects produced the more usable assets. Diffuse natural forms varied more strongly, while mass or scene-scale meanings such as dense forest could fail because they lack the boundary expected by single-object reconstruction. The same dense forest meaning failed with both reconstructors tested at that stage, but two models are not enough to claim a general limitation of image-to-3D reconstruction.

The useful distinction therefore shifted from the initial solid-versus-mass framing toward bounded versus unbounded. Tree can be represented as one object; dense forest belongs more naturally to a scene layer that this system does not provide.

Open vocabulary creates a second cost through caching. Once a meaning produces a mesh, subsequent assignments reuse it and movement does not regenerate it. The submitted interface has no ordinary reroll command, so a poor first generation becomes sticky unless the meaning is changed or the cache is altered outside the interface. The cache is therefore strong for reuse but weak for appearance-level recovery.

The recorded state nevertheless includes meanings such as windmill, fire truck, lighthouse, castle, stone bridge, gas pump and metal fence post. These do not establish an unlimited vocabulary. They establish that semantic possibilities are no longer bounded by the detector categories known when the system was built.

5.3 The two-stage generation pipeline

5.3.1 Pipeline architecture and rationale

Because semantic assignment is decoupled from physical appearance (Section 5.2.1), the reconstruction stage cannot use an image of the physical prop. A bottle cap assigned as a lighthouse would still present the reconstructor with a bottle cap. The generation path therefore

became text to image, then image to 3D. Stage 1 invents the visual form of the assigned meaning; Stage 2 reconstructs that image as geometry.

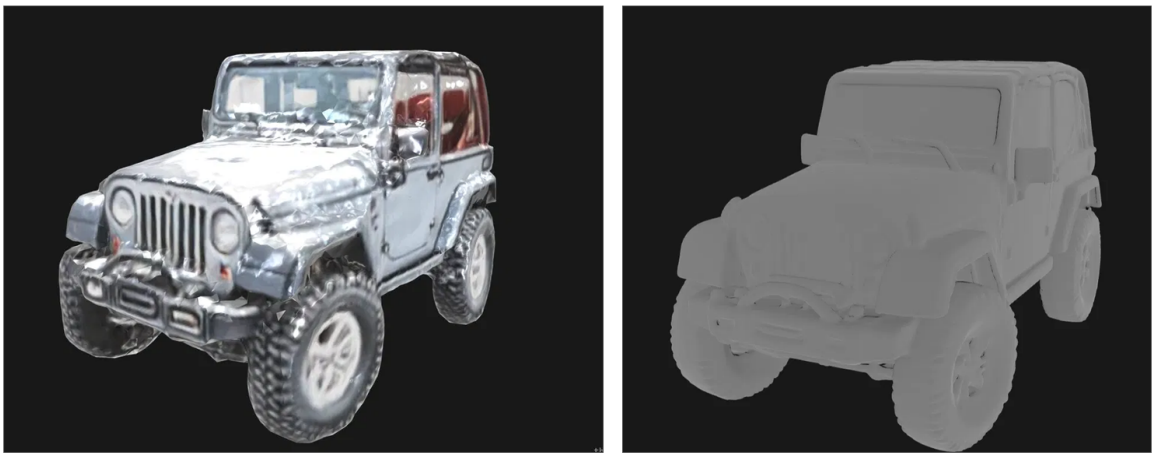
Shap-E (Jun and Nichol, 2023) was also tested as an early direct text-to-3D alternative and rejected after its outputs were recorded as bulgy and imprecise. The sequential design better matched the available hardware and, more importantly, introduced an inspectable boundary between semantic generation and reconstruction.

That boundary later became useful evidence. A defect already visible in the Stage 1 image can propagate into the final mesh, while a recognisable source image followed by severe geometric failure points to the reconstruction stage. The pipeline therefore serves not only to generate assets but also to locate where some failures enter.

5.3.2 Reconstruction model evaluation

TripoSR (Tochilkin *et al.*, 2024) was the first local reconstructor tested. It produced usable meshes, but orientation was unreliable in these tests. SF3D (Boss *et al.*, 2025) followed and was recorded as preferable for texture and orientation, with practical local runs of roughly 35–85 seconds depending on background removal.

A later comparison tested TRELLIS.2 (Xiang *et al.*, 2025), SF3D, TripoSR and Hunyuan3D-2.1 (Team Hunyuan3D *et al.*, 2025) on the same four subjects. The diary records TRELLIS.2 as the preferred output from that exploration, followed by SF3D, but this was not a formal benchmark. The decision combined visual judgement with orientation, runtime, hardware requirements and integration effort.



(a) TripoSR, textured, 6 July

(b) Hunyuan3D-2.1, shape only

Figure 5.8 Two outputs from the reconstruction-model exploration: a textured TripoSR jeep and a shape-only Hunyuan3D-2.1 jeep from different sessions. The figure records the geometry being inspected rather than a controlled comparison between the models.

TRELLIS.2 depended on a CUDA workstation in another building and therefore did not fit the live deployment requirement. Hunyuan3D eventually produced valid meshes, but required substantially more integration work than its observed improvement justified. SF3D consequently became the live path. This was a deployment decision under the tested conditions, not evidence that SF3D was objectively superior to the alternatives.

5.3.3 Failure localisation across pipeline stages

During live testing, a generated car appeared almost black and flattened in the browser. I first suspected scene lighting, but forcing midday illumination did not change it. Inspecting the

intermediate image showed a different problem: Stage 1 had produced an extreme frontal crop dominated by the grille.



(a) what Stage 1 produced for "a car" (b) the object it became, in the live scene

Figure 5.9 Black-car failure. The Stage 1 image is already an extreme frontal crop, and the resulting mesh appears dark and flattened in the Three.js scene. The framing defect is visible before reconstruction.

Three further stochastic generations of “a car” did not all become black, but each repeated the same tight frontal framing. The prompt then in use ended with “front view, centered, symmetrical, plain background”, giving a concrete reason to suspect the framing instruction.

The prompt was then changed toward a three-quarter view with the whole object visible. Stage 1 also moved from SD Turbo to SDXL Turbo (Sauer *et al.*, 2023) shortly afterwards, so the later improvement cannot be attributed cleanly to either change alone.

The important result was the debugging change: the lighting hypothesis failed, and the intermediate image revealed that a defect visible in the final 3D object could already exist before reconstruction. From that point I stopped treating a bad browser render as evidence that the renderer or reconstructor was necessarily where the failure began.

5.3.4 SPAR3D selection and remote deployment

A matched comparison reconstructed the same clean tree source image with SF3D and SPAR3D (Huang *et al.*, 2025).



Figure 5.10 The same tree source reconstructed with SF3D and SPAR3D. The SF3D result becomes extremely thin when rotated; the SPAR3D result retains a volumetric trunk and canopy. The comparison concerns this matched source image rather than general model superiority.

For this particular image, SPAR3D produced a materially more usable object. The diary also records checking it from front, back and both sides in the browser. One matched source does not support a general model ranking, but it provided enough evidence for the engineering decision to replace SF3D.

Category-specific routing between reconstructors was not introduced because it would have reintroduced a predefined semantic taxonomy into a system designed around open meanings.

The immediate cost was speed. One local SPAR3D run took 138.9 seconds of compute and 177.5 seconds wall time, compared with the 35–85 second practical range previously recorded for SF3D. That made deployment rather than local optimisation the next problem.

Both generation stages moved to a rented RTX 4090. A first cold SPAR3D request took 30.2 seconds even though its mesh-generation loop itself took 1.99 seconds, showing that startup and model loading dominated the delay. Both models were therefore converted into resident services. Once warm, the complete laptop-to-pod path returned a mesh in about two seconds; in the following battery, eight of nine source-image-to-mesh intervals were two seconds and one was three.

The local path remains as a fallback, but its Stage 1 configuration is not identical to production.

	Local script	**Pod service**
Inference steps	5	4
Device	MPS or CPU	CUDA
Model loading	Reloaded per call	Resident
Typical Stage 1 time	Seconds to tens of seconds	About 0.6 s

Table 5.3 Stage 1 configurations retained in the submitted system. The production pod uses four inference steps with resident CUDA inference; the local fallback uses five and reloads the model per call. They share the same model and handoff but are not identical execution paths.

5.3.5 Pipeline evaluation and failure modes

A subsequent thirteen-meaning battery cleared the previous outputs and tested the assembled pipeline. Seven returned clean or nearly clean assets, while six produced substantial failures. The useful result was not the overall split but where the first visible defect appeared.

Three retained failures, backpack, traffic cone and vintage motorcycle, already contained multiple instances or views in their Stage 1 images.

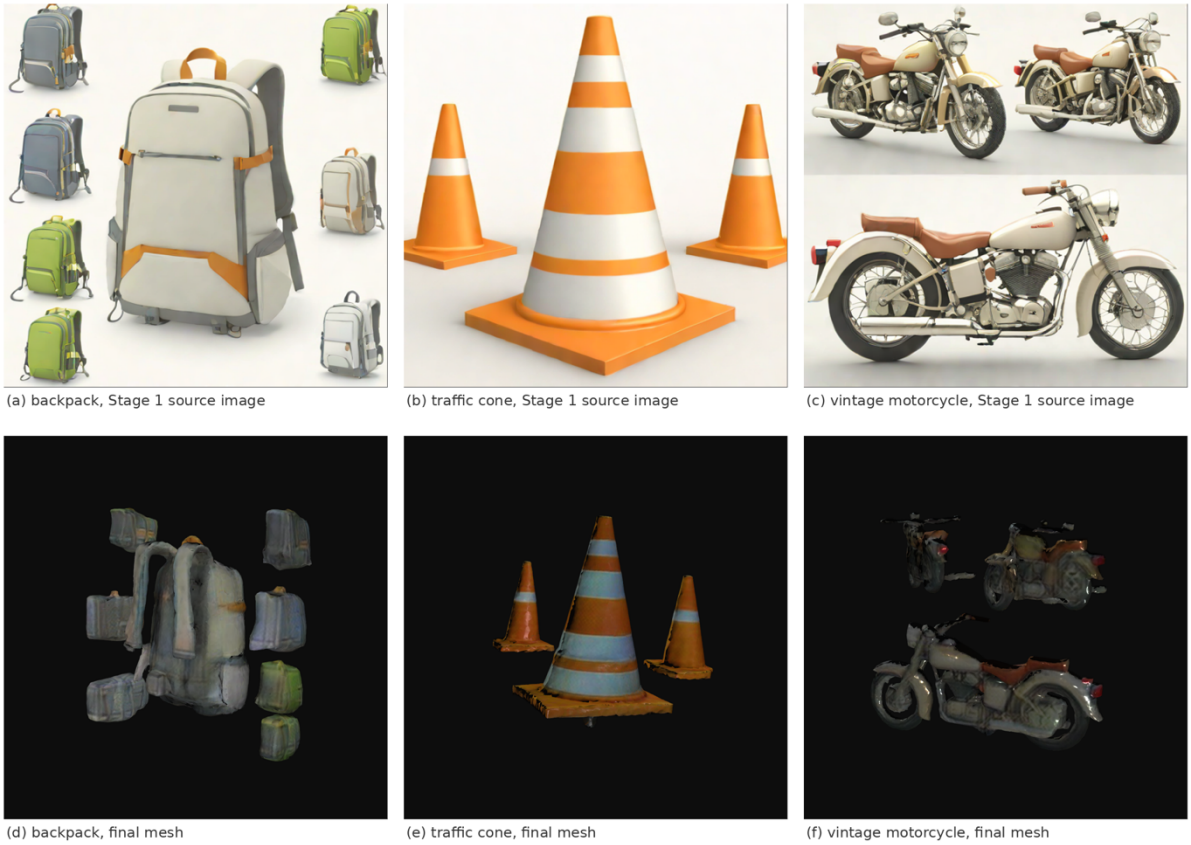
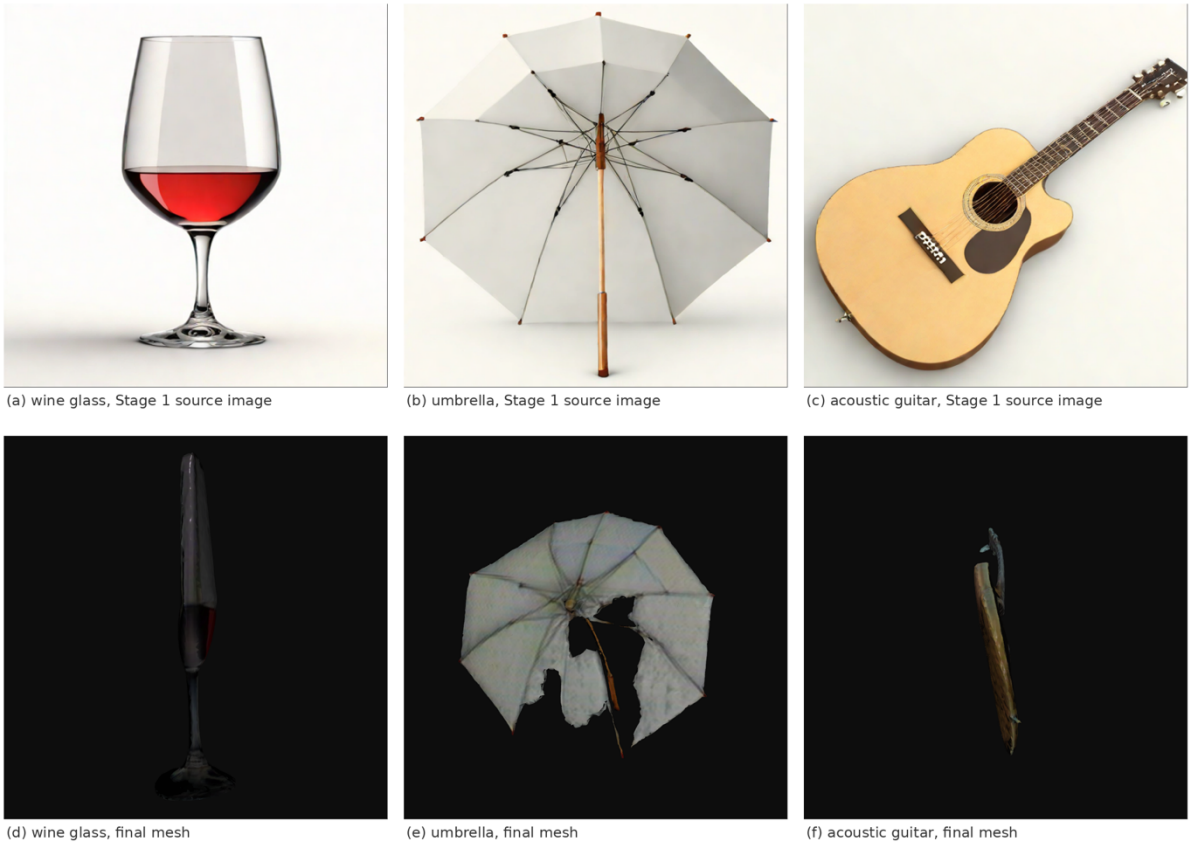


Figure 5.11 Three multi-instance failures from the thirteen-word battery. Backpack, traffic cone and vintage motorcycle already contain repeated instances or views in the images supplied to SPAR3D; the reconstructed meshes preserve that multiplicity.

For those cases, repetition entered before reconstruction. This does not establish that every multi-instance failure has the same origin, only that the preserved examples can be located on the Stage 1 side of the boundary.

Wine glass, umbrella and acoustic guitar behaved differently. Their Stage 1 images each contained one recognisable subject, but their final meshes became severely thin, incomplete or deformed.



(a) wine glass, Stage 1 source image
(b) umbrella, Stage 1 source image
(c) acoustic guitar, Stage 1 source image
(d) wine glass, final mesh
(e) umbrella, final mesh
(f) acoustic guitar, final mesh

Figure 5.12 Three single-subject failures from the same battery. The Stage 1 images contain recognisable wine glass, umbrella and guitar subjects, while the resulting meshes become thin, incomplete or heavily deformed.

I initially interpreted these failures in terms of transparency, thin structures and geometry hidden from a single view. Those remain plausible explanations, but I did not isolate them with matched tests and therefore do not claim them as general limits of single-image reconstruction.

The more defensible finding is architectural. In some retained failures the defect is already visible before reconstruction; in others the source image remains recognisable and the severe failure first appears in geometry. The two-stage pipeline therefore creates an inspectable boundary that helps locate failure without pretending to establish its underlying mechanism.

5.4 Scale control and explicit grammar

5.4.1 Scale as an authorship decision

Position already had an owner in the physical arrangement, and semantic identity had an owner in language. Scale did not. Three approaches were therefore built and measured: inference by the language model, deterministic extraction from ordinary phrasing, and an explicit reserved grammar. The first two were rejected because both could assign a size the person had not deliberately chosen.

5.4.2 Evaluation of automatic scale assignment

The first approach let Qwen classify each assigned meaning into a scale band. Three architectures were frozen and evaluated: direct enumeration, a structured version that first stated a physical basis for the decision, and a version that extracted explicit size comparisons

before classification. All used the local Qwen3-8B at temperature zero with thinking disabled and were tested against the same 72-case holdout.

Architecture	Valid contract	Acceptable overall	Ordinary concrete	Explicit modifiers	Abstractions	Boundaries	Creative
A, direct enumeration	72/72	43/72 (59.7%)	21/36	12/20	3/4	3/6	4/6
B, structured basis	72/72	50/72 (69.4%)	28/36	13/20	2/4	3/6	4/6
C, modifier pre-parser	72/72	48/72 (66.7%)	—	13/20	—	—	—

Table 5.4 Three automatic scale architectures on a frozen 72-case holdout separate from the 98-case development set. Up to two adjacent classes were accepted where defensible. All three returned valid contracts on every case; the structured version performed best overall at 50/72.

The structured version was acceptable on 50 of 72 cases. Ordinary concrete meanings were strongest at 28 of 36, but it handled only 13 of 20 meanings that already stated their own size. In seven of those cases the model overrode an explicit modifier. The failure was therefore not output formatting: every architecture returned a valid contract, but the semantic decision inside that contract could still be wrong.

A repeated-call check reinforced the problem. Twelve meanings were run three times against each candidate and produced the same classifications on every run, including two consistently wrong results. Because scale is cached with meaning, a confident mistake could persist for the session rather than disappear through variation.

The second approach avoided model inference and instead tried to read size directly from ordinary wording. A deterministic parser required one of six size words at the start of the phrase, optionally after “a” or “an”, stripped that word from the generation meaning, and applied the corresponding target. On a frozen 60-case holdout it reached 72.7% precision and 66.7% recall.

Its nine false positives exposed the limit. “Big Ben”, “large language model”, “small talk” and “miniature painting” all contain apparent size words whose meanings are not scale instructions. Across an open vocabulary, a parser cannot safely assume that such words are control rather than semantics. The same test confirmed that explicit scale values could be applied and persisted correctly; only automatic detection was rejected.

5.4.3 Explicit scale grammar

The third approach stopped guessing. Scale became an explicitly marked control using scale <size>.

Two marked grammars, scale <size> and make it <size>, both separated control from ordinary meaning without error in an initial 90-case holdout. scale was retained because it directly names the controlled property and avoids collision with atmosphere commands such as “make it foggy”. A 120-case follow-up also showed that the comma before the marker could be optional.

Three marker forms were then compared on a fresh 148-case semantic holdout.

Marker	True positives	False positives	Precision	Recall
size <value>	24	3	88.9%	100%
size is <value>	24	6	80.0%	100%
scale <value>	24	0	100%	100%

Table 5.5 Three scale-marker candidates on a fresh 148-case semantic holdout. size and size is produced false positives on ordinary descriptions; scale handled all tested positive cases without altering any negative case.

The alternatives failed on ordinary phrases such as “a T-shirt size small” and sentences ending “whose size is huge”. Scale produced no false positives in this holdout. That does not establish that the word could never occur as ordinary meaning, but it was the safest of the tested markers.

5.4.4 Renderer scale calibration

The production grammar maps six words onto five visual targets: tiny and miniature to 0.35, small to 0.65, no marker to 1.00, big and large to 1.80, and huge to 3.00. These are relative visual sizes rather than physical dimensions.

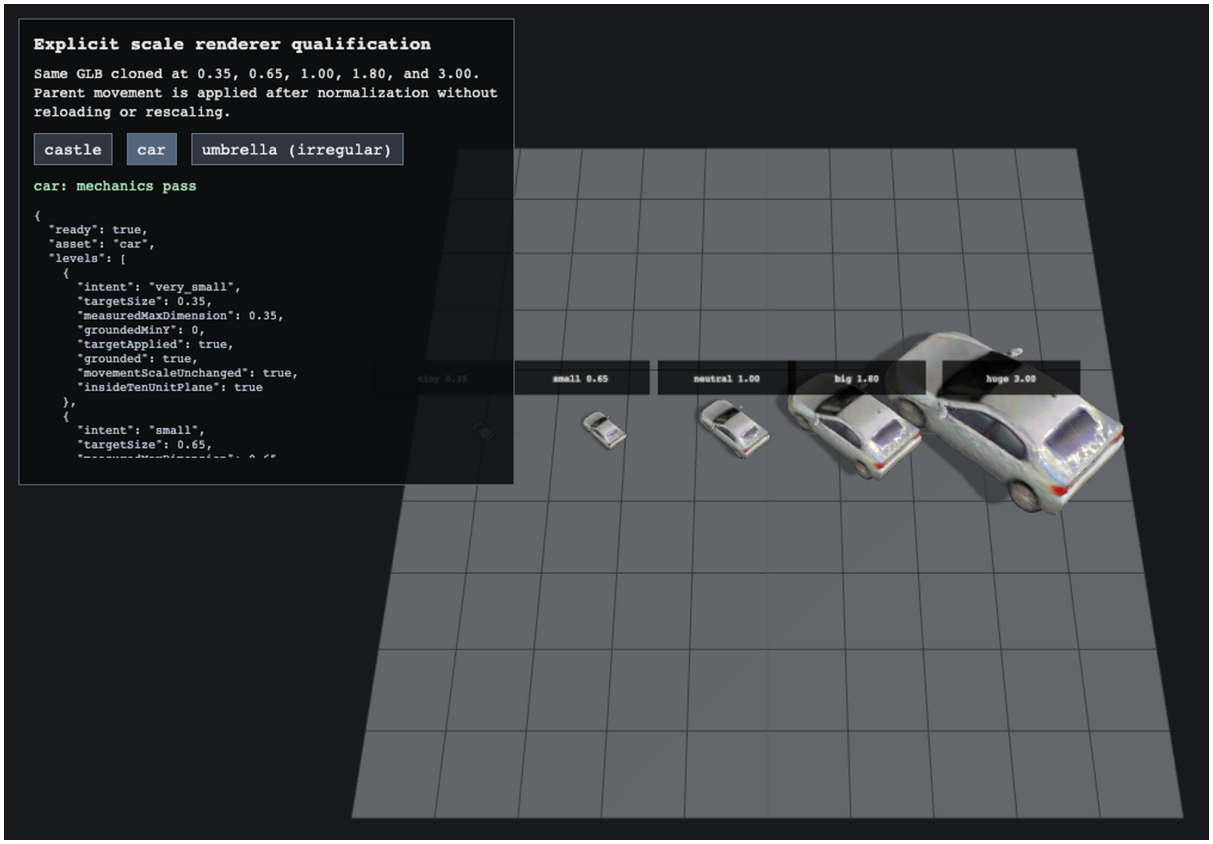


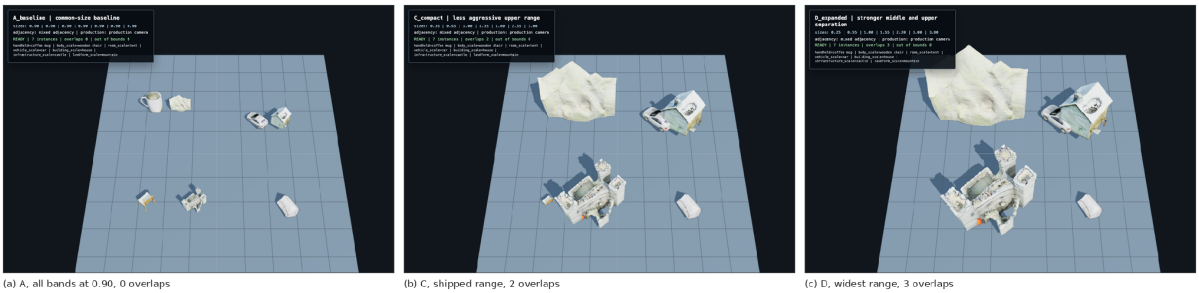
Figure 5.13 Renderer qualification for the five production targets using the same car mesh at 0.35, 0.65, 1.00, 1.80 and 3.00. The checks verify target dimension, grounding, persistence after movement and table bounds. Castle and umbrella meshes were tested through the same qualification.

The targets came from an earlier seven-band calibration comparing five candidate ranges under adjacency, crowding, edge stress and a visual reference condition.

Candidate	Seven values	Adjacency overlaps	Crowding overlaps	Edge overlaps	Edge out of bounds
A, common-size baseline	0.90 × 7	0	0	0	0
B, current manual	0.30, 0.60, 1.00, 1.40, 2.00, 2.70, 3.40	2	0	0	1
C, less aggressive upper range	0.35, 0.65, 1.00, 1.35, 1.80, 2.35, 3.00	2	0	0	1
D, stronger middle and upper separation	0.25, 0.55, 1.00, 1.55, 2.20, 3.00, 3.80	3	0	1	1
E, balanced alternative	0.32, 0.62, 1.00, 1.42, 1.95, 2.50, 3.10	2	0	0	1

Table 5.6 Five seven-band candidate ranges tested across adjacency, crowding, edge stress and a visual-reference condition, with 59 of 60 planned frames surviving. The original counting script is unavailable, so the archive does not establish whether "overlap" referred to 3D geometry or rendered-image overlap.

The constant-size control, candidate A, recorded no collisions because it removed meaningful size variation altogether. Candidate D, the widest spread, produced the most recorded collisions. B, C and E tied on the retained quantitative measures.



(a) A, all bands at 0.90, 0 overlaps (b) C, shipped range, 2 overlaps (c) D, widest range, 3 overlaps

Figure 5.14 Three candidate ranges under the same adjacency condition. Candidate A removes scale variation, candidate C preserves a moderate range, and the wider candidate D produces stronger spatial interference and makes its smallest object visually negligible.

The five shipped targets correspond to C's first, second, third, fifth and seventh values. The archive does not record why C was selected from the three quantitatively tied candidates, so the calibration supports a bounded working range rather than a claim that the final values were optimal.

5.4.5 Speech-recognition integration

The explicit grammar then met a probabilistic input channel. Spoken “object two is a car scale tiny” repeatedly returned as “a car scaled tiny”. Because scaled is not the reserved marker, the parser correctly treated the entire phrase as ordinary meaning, creating a different cache identity and generating another car rather than applying the requested scale.

The obvious repair was to accept scaled <size> as an alias everywhere. That recognised the mistranscriptions but also produced six false positives on ordinary descriptions such as “a VFX asset already scaled tiny” and “a source mesh uniformly scaled big”. A global alias would therefore recreate the same silent reinterpretation that had already disqualified adjective parsing.

The accepted repair was narrower. The alias runs only after a voice transcript has matched the assignment structure object N is ..., only when scaled <size> appears at the end, and only for voice. Typed “a car scaled tiny” remains ordinary semantic input. In the declared-grammar holdout, all 36 alias cases and all 68 non-alias cases were handled correctly.

The cost is deliberate. A spoken meaning genuinely ending in “scaled tiny” cannot be expressed literally through that grammar and must be rephrased. The system gains a reliable control boundary by reserving a narrow region of spoken language rather than trying to infer which interpretation was intended.

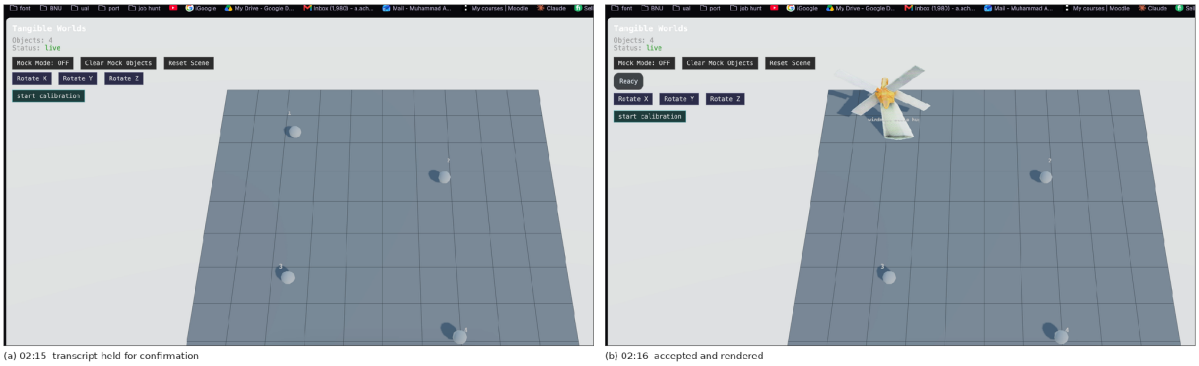


Figure 5.15 Spoken assignment during the final voice implementation. Before acceptance, four numbered placeholders remain unchanged; after confirmation, the windmill appears at target 3.00. The recorded transcript was “object one is a windmill scale huge”; scale was recognised correctly here, so the voice alias was not used.

The original scaled experiments used Moonshine before the production recogniser was replaced. That recogniser change is discussed in Section 5.5.

5.4.6 Design implications for scale control

Scale is where the division of labour had to be designed rather than inherited. Position comes from physical arrangement and semantic identity from language, but size could plausibly have been inferred automatically. Both model classification and ordinary-language parsing failed for the same reason: they could produce a scale the person had not explicitly assigned.

The reserved marker adds interaction cost. It must be learned, used in a particular position, and a narrow phrase range is unavailable as ordinary spoken meaning. In return, size is not silently inferred from semantics. The broader lesson is not that explicit syntax is always preferable to natural language, but that a property presented as part of a person’s authored

scene should not be assigned implicitly when the system cannot reliably distinguish instruction from meaning.

5.5 Voice input and confirmation

5.5.1 Voice input and semantic uncertainty

The abandoned study explains why the interface still accepts a condition and participant_id, but voice remains relevant because it introduces probabilistic recognition between linguistic intent and scene state. Section 5.4.5 showed one consequence: a valid scale instruction could be transcribed as a different semantic phrase. The broader design problem was therefore how to expose recognition uncertainty before a transcript could modify the scene.

5.5.2 Speech-recogniser evaluation

The initial recogniser bake-off used 30 microphone recordings covering connected speech, command vocabulary, long pauses and adversarial inputs. Nine faster-whisper checkpoints, based on Whisper (Radford *et al.*, 2023), and Moonshine (Jeffries *et al.*, 2024) were each tested with and without decoder hints, producing 20 configurations and 600 transcriptions.

Checkpoint	Critical-token survival, 23 command clips	Adversarial clips left clean, of 3	Mean transcription time	Model load time
moonshine-en	96.2%	2	0.12 s	17.3 s
large-v3	96.2%	0	5.97 s	182.0 s
large-v3-turbo	94.2%	0	4.58 s	96.0 s

Table 5.7 Selected results from the initial speech-recognition bake-off. Moonshine and Whisper large-v3 both retained 96.2% of critical command tokens in the tested command clips, but differed substantially in latency, model loading and behaviour on adversarial audio.

The adversarial clips changed the decision. With command-vocabulary hints enabled, large-v3 turned silence into plausible atmosphere phrases such as “stormy lightning rain”. Without hints it hallucinated generic speech instead. Moonshine returned empty transcripts on two of the three adversarial clips and only a short fragment on the third. Decoder hints were therefore rejected because they made false commands more plausible precisely when no command had been spoken.

Moonshine was selected for the first production path because it combined strong critical-token survival in this test with much lower latency and a safer tested failure profile on non-command audio.

5.5.3 Confirmation and bounded repair

Recognition alone is not allowed to change the world. Voice input is triggered by holding the DualSense touchpad; an AudioWorklet captures 16 kHz mono audio and sends it to the local recogniser. The transcript is then read back through local TTS and must be accepted before application.

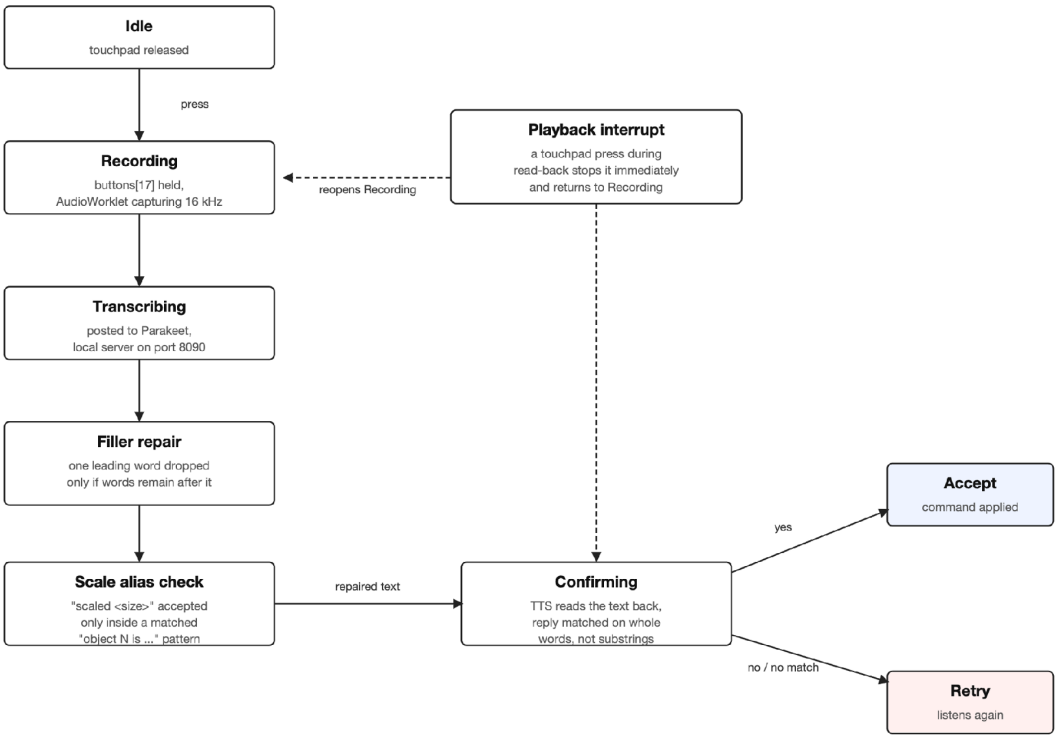
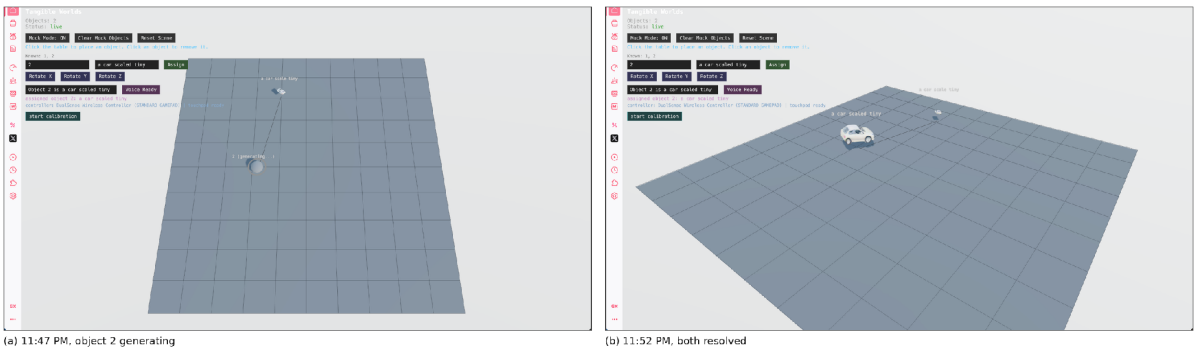


Figure 5.16 Push-to-talk and confirmation state machine. Speech is transcribed, bounded repairs and the voice-only scale alias are applied, then the result is read back for acceptance or retry. A touchpad press during playback interrupts the read-back and immediately reopens recording.

Confirmation uses whole-word matching rather than substrings, avoiding collisions such as no inside know. A touchpad press during playback interrupts TTS and returns directly to recording (Matsuyama *et al.*, 2010; Lin *et al.*, 2022). Rejected transcripts remain in the interaction log rather than disappearing.

A separate failure involved leading filler words. The initial theory was that dead air before speech caused the recogniser to invent an opening word, but the measurements contradicted it: clips with junk words averaged 0.38 seconds of leading silence, compared with 0.66 seconds for clean clips.

A 50-clip trimming test also failed to justify preprocessing. Leading-junk counts did not improve reliably, while valid commands could be damaged: “Add lightning” became “had lightning” whenever trimming was applied. Audio trimming was therefore dropped, and repair was limited to removing one recognised leading filler only when meaningful words remained afterwards.



(a) 11:47 PM, object 2 generating

(b) 11:52 PM, both resolved

Figure 5.17 The scale-grammar failure in use. Typed a car scale tiny produces the intended small car; the spoken version, transcribed as a car scaled tiny, is treated as a different ordinary meaning and renders at neutral scale.

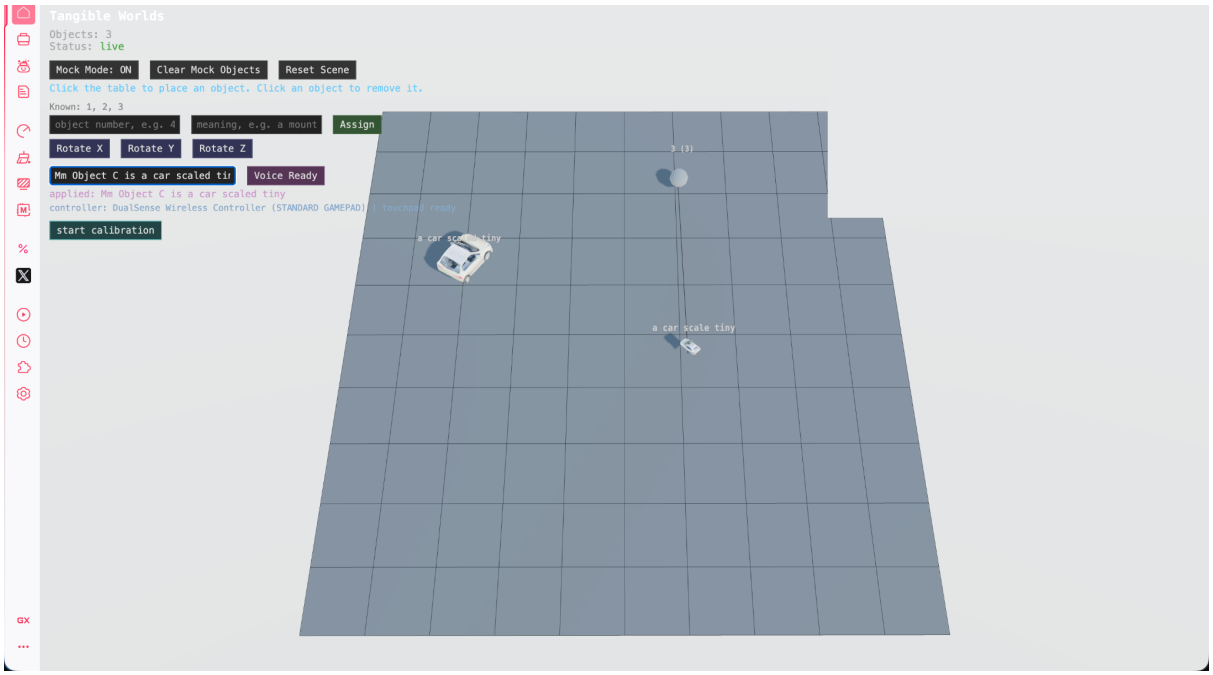


Figure 5.18 Multiple recognition errors in one transcript: a leading filler, 3 heard as C, and scale heard as scaled. The frame is matched to the interaction log from the same minute.

5.5.4 Interaction timing and retry evidence

The typed interface remains deliberately simpler: the person enters a meaning, presses Assign and sees what was applied.

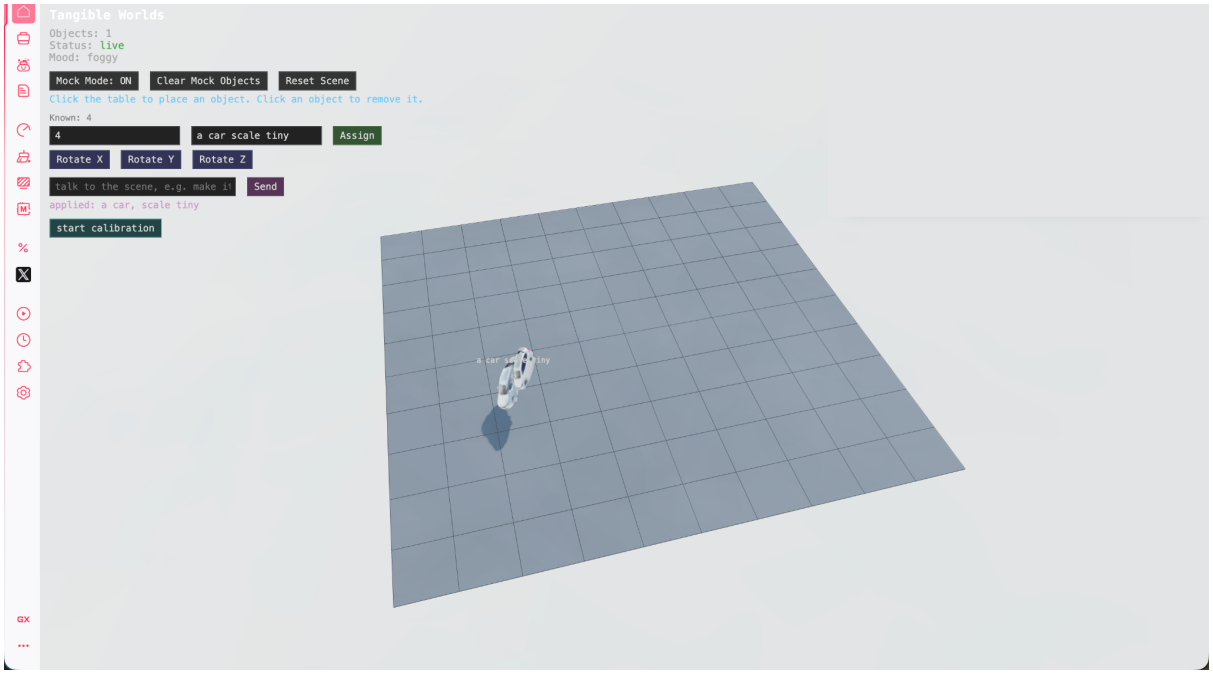


Figure 5.19 Typed input path. Meaning is entered and assigned directly, without transcription, spoken confirmation or retry handling.

This simpler path also provides a useful comparison point for the interaction timings recorded in the development log.

	Voice, with a full timing record	Typed
Entries	18	12
total_command_s range	13.4 – 56.9 s	8.6 – 34.8 s
total_command_s mean	27.4 s	13.5 s

Table 5.8 Completion times for timed development-log entries: 18 voice and 12 typed interactions. The entries were not matched by command, state or retry count, so the values describe recorded interaction cost rather than a controlled modality comparison.

Voice entries averaged 27.4 seconds and typed entries 13.5 seconds, but those figures should not be interpreted as a clean estimate of a "voice penalty". Voice performs speech recognition and confirmation in addition to the model call, and the logged commands were not paired.

Retries provide a more direct record of recognition friction. Across 36 logged voice interactions, 27 required no retry, four required one, two required two, two required four, and one required eight. The eight-retry case is therefore the tail rather than the typical recorded interaction.

5.5.5 Production recogniser re-evaluation

Moonshine was the first production recogniser. After repeated recognition problems, that decision was reopened and Parakeet TDT 0.6B v3 (NVIDIA, 2025) was qualified on a 102-clip corpus covering assignment commands, robustness cases, atmosphere, confirmation and adversarial audio.

Corpus category	Clips
Assignment core	36
Assignment robustness	12
Atmosphere	16
Clear confirmation	20
Ambiguous confirmation	6
Adversarial non-command audio	12
Total	102

Table 5.9 Composition of the 102-clip Parakeet qualification corpus. All requests completed successfully and 89/102 transcripts exactly matched their reference text. The surviving result file does not preserve category-level exact-match counts.

The 13 non-exact transcripts varied in consequence. Some were trivial differences such as nah becoming na. Others could damage routing: “make it cooler” collapsed into unrelated Cyrillic text, while “a market stall scale big” became “scale bag”. The latter fails the reserved grammar rather than silently applying another scale, which is preferable but still requires recovery.

Median transcription latency was 43.3 ms and the 95th percentile 55.7 ms. A separate 13-clip Qwen3-ASR (Shi *et al.*, 2026) probe produced a routing-damaging substitution, “Object three” becoming “Objective”. Because it was tested on a different and much smaller corpus, the Qwen3-ASR probe is not treated as a direct numerical comparison with the 102-clip Parakeet result.

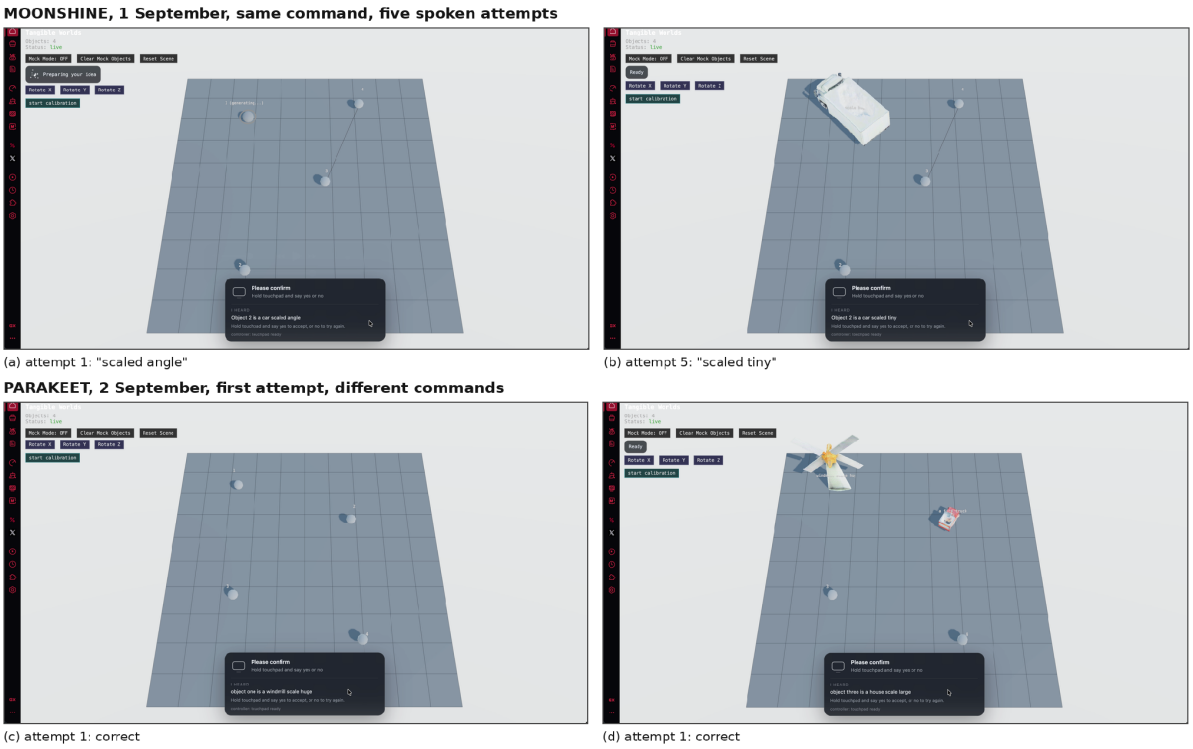


Figure 5.20 Illustrative interaction evidence from the recogniser transition. A Moonshine scale assignment required repeated attempts, while two later Parakeet assignments were accepted first time. The recordings and commands differ, so the figure illustrates the change in use rather than isolating recogniser performance.

Parakeet TDT 0.6B v3 through NeMo-Speech.cpp replaced Moonshine in the production system.

5.5.6 Design implications for voice input

The central finding is not that one recogniser was sufficiently accurate. It is that once speech can author world state, recognition uncertainty cannot be allowed to become state silently.

The final voice path therefore combines confirmation before application, explicit scale syntax, narrowly bounded transcript repair, a voice-specific alias for one measured recognition error, and retention of rejected attempts. These mechanisms do not make speech recognition correct. They make its uncertainty visible and recoverable before it acquires authority over the scene.

Changing from Moonshine to Parakeet improved the production recogniser but did not remove that requirement. Parakeet can still corrupt scale markers and other command tokens. Confirmation therefore remains part of the interaction architecture rather than a workaround for one model.

5.6 Atmosphere and environment

5.6.1 Initial atmosphere implementation and failure

Atmosphere was built parametrically rather than as a generated skybox, prioritising responsiveness and reliability over another generative stage. Language reaches the renderer through /api/converse. An earlier implementation used Ollama and Llama 3.2; the submitted system uses local Qwen3-8B.

That first version was hardware-tested and failed visibly. "Golden hour" produced a warm sky over a near-black ground. "Add mist" was barely visible. "Spooky" collapsed the whole scene toward unreadable black.

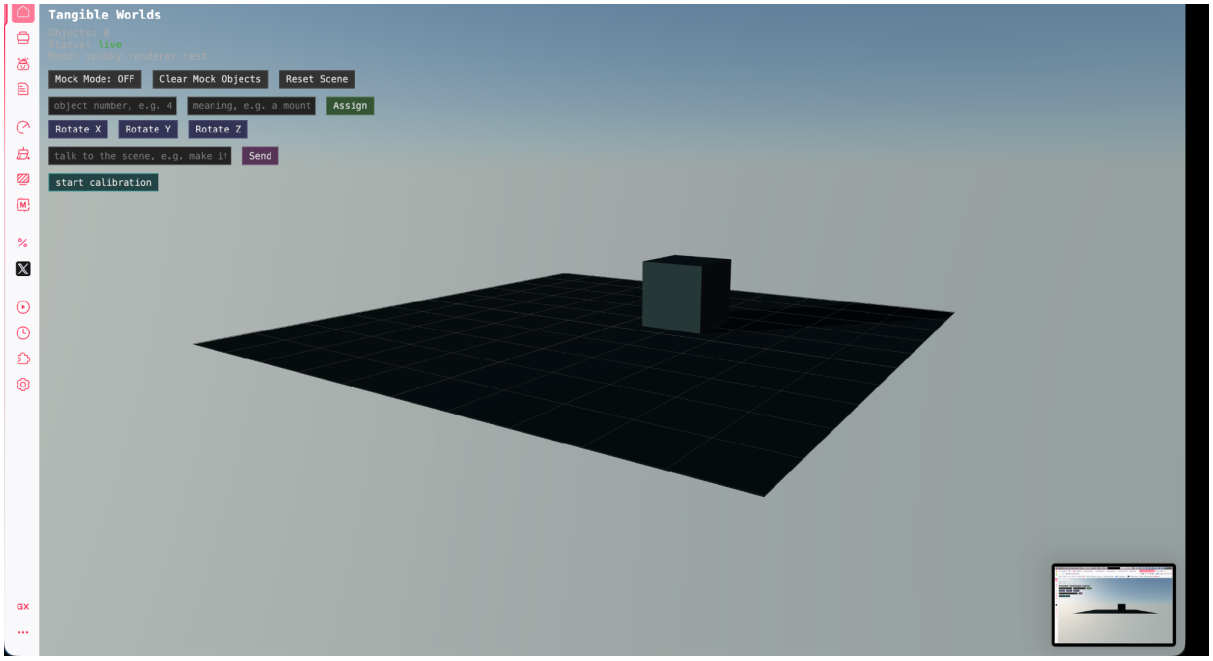


Figure 5.21 A diagnostic renderer test, the mood field set directly to "spooky renderer test" rather than reached through language, to isolate whether the failure was in the model or the renderer. The table reads almost black under a flat grey sky; a single cube is barely distinguishable from the ground it sits on.

The log located the problem more clearly than the screenshot. Across all commands recorded before the rebuild, ground_color remained #2a2a2a. The language model was therefore not changing that field, indicating a control-path omission rather than a shading failure.

5.6.2 Parametric atmosphere representation

The rebuild, Phase 5E, replaced direct control of colours and light vectors with twelve semantic fields: eleven numeric render parameters and a free-text mood string. The renderer maps the numeric values deterministically to lighting, sky, mist and weather; mood is retained only for logging and display.

The renderer also moved away from camera-distance fog and billboard mist toward world-space procedural mist. A generated skybox was considered but not adopted, preserving the parametric architecture rather than introducing another generative stage.

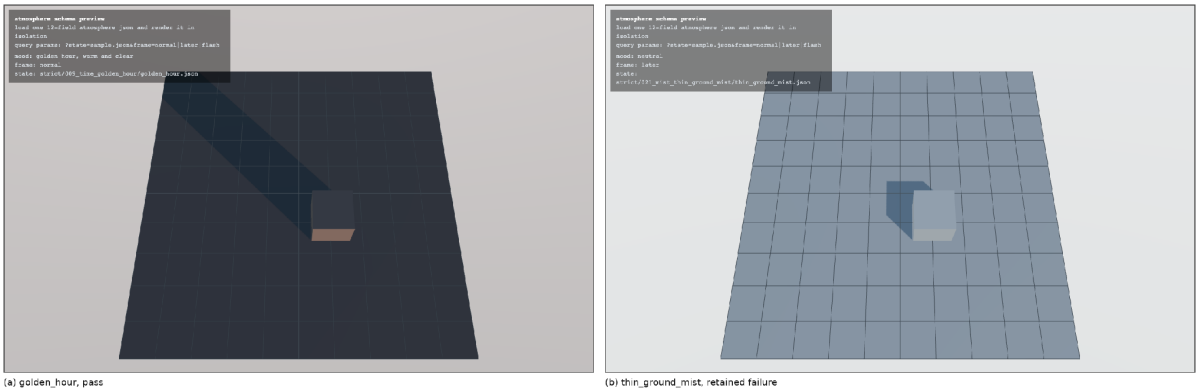
5.6.3 Qualification results and persistent failures

A 68-case qualification battery covering time of day, lighting, mist, weather, conflicting instructions and open vocabulary was run twice. The gate required at least 90% across the strict, sequence and repeated-call tests, all ten preservation cases to pass, and zero schema failures. The first run scored 59/68 strict cases and failed; the second scored 62/68 and passed.

	First run	**Second run**
Strict cases (68, incl. 10 preservation)	59	62
Explicit preservation	10/10	10/10
Sequences, 3×5 steps	14/15	15/15
Repeated calls meeting case criteria, 5×3	15/15	15/15
Schema failures	0	0
Gate	fail	pass

Table 5.10 Two qualification runs using the same strict cases, sequence steps, repeated-call cases and semantic validation criteria. The later Playwright harness additionally captured visual evidence for all 106 model calls, with no screenshot failures. A repeated call passed when it satisfied that case's expected ranges and preservation checks; this row does not measure equality between repeated outputs.

Six cases failed both runs: early_morning, thin_ground_mist, fast_drifting_mist, violent_thunderstorm, stormy_no_precipitation and warm_night. Their common pattern was partial execution of multi-field instructions. In warm_night, light_warmth moved from 0.0 to 1.0 while sky_elevation remained at its daytime default. In fast_drifting_mist, mist_motion reached its maximum while mist_amount remained zero. Most starkly, stormy_no_precipitation changed no numeric field at all; only the mood string became “stormy dark clouds”. A plausible display label therefore did not establish that the rendered state had followed the command.



(a) golden_hour, pass (b) thin_ground_mist, retained failure

Figure 5.22 The qualification harness's own debug view, which states which state file it loaded rather than leaving that to be inferred. (a) golden_hour, passing: a raking shadow across the table, confirming that the light angle moved rather than only its colour. (b) thin_ground_mist, failing: mist_amount moved to 0.10, inside its accepted range, but mist_height remained at its 0.35 default rather than entering the required 0–0.30 ground-hugging range. The rendered frame remained visually faint enough that little mist is apparent.

Ninety-one percent of the battery passed. The persistent failures clustered around multi-field instructions in which one requested parameter changed while another remained at default.

5.6.4 Repeatability and design implications

Two separate checks concern repeatability, and they should not be collapsed into one metric. The 15/15 repeated-call row in 5.10 means that every call independently satisfied its case's expected ranges and preservation rules. It does not test whether repeated outputs were identical. The saved JSON was therefore checked separately. In the second run, all five three-call sets were byte-identical within each case, including all eleven numeric render parameters and the mood string. The separate rendered-image comparison described in Section 3 is a different check, operating on rendered frames rather than JSON state. These findings agree, but they operate at different layers.

The rendered frames were compared as well. Across the five commands, 461 pixels differed out of 5,120,000 compared, and every difference fell within ten rows near the top of the frame. Outside that band the three runs of each command were pixel identical.

The more specific lesson concerns where language control is evaluated. A command can update the mood label correctly while leaving the numeric render state unchanged. Validation therefore needs to inspect the eleven render parameters rather than infer success from the display label.

5.7 Terrain and interstitial relations

5.7.1 Object-anchored terrain design

The first terrain experiment treated the ground as one environment-wide surface controlled through language. That was spatially wrong for this project: a generated mountain or river could change, but its location no longer came from a physical arrangement.

The terrain design therefore moved to object-anchored procedural terrain. A prop centre set the landform position, its measured footprint set the radius, and measured height set the amplitude. Positive fields such as hills and mountains raised the surface; negative fields such as valleys and rivers lowered it. Because the fields combined additively, overlapping positive and negative forms could create effects such as a channel through a mountain without a specific pair rule. Moving a prop immediately moved its contribution to the surface.

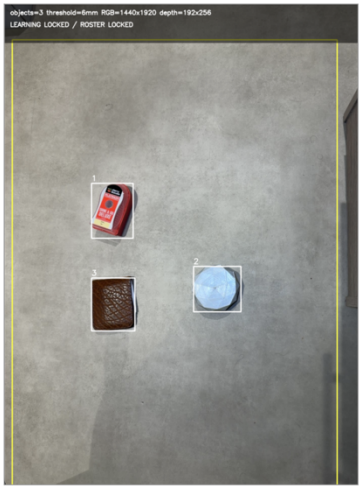
MESA (Borne-Pons *et al.*, 2025) was also tested as a learned alternative. Qwen-to-MESA prompting produced aligned RGB and elevation outputs, and several rugged terrain examples were visually promising, but geometry varied substantially across the morphology classes tested. Convincing RGB appearance also did not guarantee a useful elevation field. MESA therefore remained comparative evidence rather than the production geometry source.

Direction	What worked	What remained unresolved	Final status
MESA terrain	Qwen-to-MESA prompting worked; aligned RGB and DEM were generated; tested rugged cases were promising	Geometry was inconsistent across the tested morphology classes	DEM rejected for production geometry
Object-anchored procedural terrain	Physical position and footprint drove local landforms; fields combined without pair rules; one live session met all six stated test criteria	Handled height was unstable; terrain remained visually small; production still required ground-height integration	Prototype only; not integrated into production.
Interstitial relations	End-to-end relation asset generation and midpoint rendering worked; a later bounded-connector case rendered successfully	Relation choice had to remain representable as a bounded object; scale, placement and orientation could still fail	Retained in the implemented production path

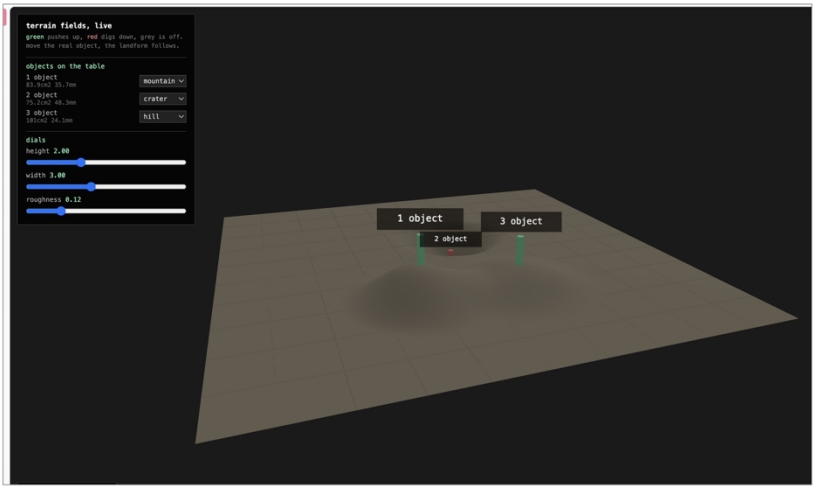
Table 5.11 Three late spatial directions and their final status. "Worked" records behaviour demonstrated in the corresponding prototype or implementation rather than reliability across arbitrary inputs or validation with users.

5.7.2 Terrain evaluation and integration limits

A live test of terrain_live.html was run after the tracker had been frozen. Three physical props-controlled mountain, crater and hill fields. The session met all six stated criteria: movement while retaining meaning, hand occlusion, merge and split, removal and replacement, field blending without pair-specific rules, and flat ground when no field was present. These results remain subject to the identity-recovery limits already established in Section 5.1.



(a) 20 Aug live terrain test, physical props



(b) 20 Aug terrain_live.html result

Figure 5.23 Live object-anchored terrain test. Three physical props control mountain, crater and hill fields. The session met the six stated test criteria, subject to the tracker limitations in Section 5.1; the fixed test vocabulary does not demonstrate open semantic terrain assignment.

Because the prototype used fixed terrain meanings rather than the open assignment path from Section 5.2, the test establishes physical anchoring, field interaction and identity continuity, not unrestricted language-to-terrain integration.

The session also exposed the main barrier to production integration. The red prop measured 36 mm early in the run and 55 mm after handling despite its physical shape being unchanged; the wallet similarly returned shorter after removal and replacement. Footprint was comparatively stable, but height was not. Because height-controlled amplitude, handling the same prop could materially change the relief it authored.

There was a second limitation at the rendering layer. The landforms occupied only a modest part of the plane and read more as bumps than as landscape. More importantly, the production renderer still uses a flat PlaneGeometry, with no displaced-ground height lookup for generated objects. Integrating terrain would therefore require both a more stable physical amplitude control and a new object-grounding path.

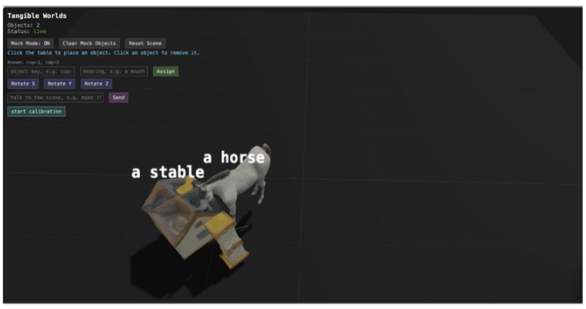
The prototype nevertheless established the underlying spatial mapping, so I treat it as an unfinished but informative direction rather than a failed experiment.

5.7.3 Interstitial relation generation and representability

The interstitial strand explored a different consequence of physical proximity: two assigned objects could trigger the generation of a third object representing a relation between them.

The first full test exposed a distinction between asset generation and spatial presentation. Horse and stable produced a recognisable wooden hitching post, but the connector was too large for the gap, intersected the stable and was poorly oriented. The generated asset was usable, but its spatial presentation was not.

Mountain and tent exposed a different problem. The original relation prompt produced a hiking trail, which is a landscape feature rather than a bounded object that the existing generation pipeline could place between two handles. The example was changed to a stone cairn, with path, trail and road outputs explicitly excluded. The next run produced a bounded cairn that could pass through the existing object pipeline.



(a) horse + stable, first spatial failure



(b) wooden hitching post, Stage 1 source image



(c) mountain + tent + cairn, improved case



(d) stone cairn, Stage 1 source image

Figure 5.24 Two interstitial tests. Horse and stable produce a recognisable hitching-post asset whose scale and placement fail in the scene. Mountain and tent produce a bounded cairn after the prompt is revised away from landscape-like connectors. Because stone cairn was supplied as the example, the latter tests representability and placement rather than independent relational inference.

Because stone cairn was supplied as the example, the result does not demonstrate that Qwen independently inferred an appropriate relationship. It demonstrates only that, once the connector was constrained to a representable discrete object, the downstream generation and placement path could handle it.

Unlike terrain, the relation mechanic remains in the implemented production system. When a ready pair enters the proximity condition, /api/relate can ask Qwen for a connector, generate or reuse its mesh, and render it between the pair. The production renderer uses a relation target size of 0.65 at table level. These changes address the known presentation failure, but the project does not establish the quality of generated relations across arbitrary object pairs.

Across these strands, the recurring issue is not only whether a mechanism works, but which scene property it allows the person, the interface or the generative model to determine. Section 6 draws those cross-strand implications together.

6 Discussion

6.1 Legibility as traceability of authored intent

The development process changed what I understood by a legible mapping. Here I use legibility in an architectural rather than perceptual sense: whether an input has a traceable relationship to the state that follows from it. In Urp and the reacTable, much of that relationship is stabilised because tangible objects already carry defined computational meanings (Underkoffler and Ishii, 1999; Jordà *et al.*, 2007). Tangible Worlds deliberately removes that certainty. Its physical handles are semantically neutral, while meaning and generated appearance enter downstream through language and probabilistic models. Open generative semantics therefore changes the mapping problem. Accuracy alone is insufficient if an error silently changes what the person appears to have authored.

The same issue emerged across unrelated mechanisms. Tracking prefers visible refusal to silent identity transfer; open vocabulary avoids a whitelist deciding available meanings; prompt shaping may frame but not elaborate the subject; and spoken input is confirmed before it can modify the world. Each exposes uncertainty before it becomes consequential state.

This extends the fixed mappings discussed in Section 2 with a different requirement: once semantic interpretation is assigned at runtime, legibility also depends on where uncertainty can resolve automatically. Kocielnik, Amershi and Bennett (2019) similarly show that the character of AI errors, not only aggregate accuracy, can shape how imperfect systems are perceived. Tangible Worlds narrows this to authorship: an error is particularly consequential when it crosses a boundary the interaction implied belonged to the person.

The resulting design introduces concrete interaction costs. Refusal can interrupt tracking, an unsuitable generated meaning may require rephrasing, and speech adds a confirmation turn. These are accepted costs of keeping uncertain interpretation exposed in the interaction rather than allowing it to become scene state without an explicit opportunity for correction.

6.2 Authorship is distributed property by property

Section 1 defined spatial authorship as authorship exercised through arranging and revising spatial structure. The artefact implements that directly. Physical handles determine position and proximity while their assigned meanings persist through movement. This differs from the predefined functions of Urp and the reacTable, but also from Tangible Diffusion, where prompts, models and tangible elements are substantially prepared in advance (Han, Tang and Wang, 2024). Making the physical handle semantically neutral moves that commitment from system design into runtime language. The cost is that a previously stable tangible mapping now depends on semantic interpretation and generation.

Authorship is therefore not transferred wholesale from model to hands. It is distributed across properties. The hands determine spatial arrangement. Language supplies semantic identity and world-level conditions. The generation pipeline determines much of the exact visual form. The person authors "mountain", but not the precise mountain produced by the first stochastic generation. Once generated, however, that appearance becomes sticky because the cache reuses the mesh for the same meaning. The architecture consequently distinguishes semantic authorship from visual determination rather than treating generation as transparent execution of a complete specification.

Scale shows why these boundaries cannot simply follow modality. The best automatic Qwen design produced an acceptable classification for 50 of 72 cases in the fresh holdout, while adjective parsing could reinterpret ordinary meanings such as “Big Ben” as control. The reserved scale <size> grammar therefore makes the decision explicit. What mattered was whether scale could be assigned without silently taking a decision the interaction implied belonged to the person.

Interstitial relations expose the opposite case. Two physical objects establish the condition that a relation exists, but the implemented mechanism deliberately delegates the connector concept to the language model. The person authors the anchors and their proximity; the system is permitted semantic latitude over what appears between them. The experiments do not establish that those choices are reliably appropriate, and the later cairn case was prompt-conditioned rather than independent inference. They do establish a different distribution of responsibility.

Taken together, these cases suggest that delegation itself is not the problem. Tangible Worlds deliberately delegates exact visual appearance and, for interstitials, some connector semantics. The failures that repeatedly caused redesign were cases of undeclared delegation, including failures through which an automatic layer effectively acquired authority over a decision that the interaction presented as belonging elsewhere. Prompt elaboration, scale inference and adjective parsing did this directly; an identity-transfer error could do it indirectly by moving an already authored meaning onto the wrong physical handle.

6.3 Exploration through independent revision

This division also clarifies the “creative exploration” part of the research question. The tangible-manipulation findings discussed in Section 2 cannot be extended behaviourally here because no creator sessions were conducted. Tangible Worlds instead establishes an architecture in which spatial and semantic dimensions can be revised independently, providing a basis on which such behaviour could later be studied.

Creativity-support literature provides a narrower basis for interpreting that architectural possibility. Shneiderman (2007) describes support for creative exploration in terms that include generating alternatives, incremental and reversible interaction, and the ability to return to earlier states. Tangible Worlds supports some of these conditions: position, assigned meaning, scale and atmosphere can be revised independently. It supports others less well. Generated appearance becomes persistent once cached, and the submitted interface provides neither an ordinary reroll command nor a history mechanism for returning to earlier visual alternatives. Shneiderman’s criteria therefore help distinguish architectural support for revision from evidence that creative exploration actually occurred.

A handle can move without changing its meaning, or be reassigned without requiring a different prop. Separate handles can share the same assigned meaning while retaining distinct object identities and positions. Scale and atmosphere can change without changing the arrangement, while proximity can introduce or remove a relational condition. These operations expose scene dimensions to separate revision rather than requiring the whole environment to be redescribed.

The separation is incomplete. Appearance-level exploration is poorly supported. Object generation is stochastic only when a meaning is first generated; after that, the cache reuses its mesh, movement does not regenerate it, and the submitted interface provides no ordinary

reroll command. Semantic and spatial alternatives are therefore directly available for independent revision, while alternative appearances for one retained meaning are not. “Creative exploration” remains an afforded possibility rather than an observed effect, and even that possibility is uneven across the properties the system exposes.

Terrain marks another boundary. It attempted to extend physical authorship from placing discrete objects to changing continuous ground. The problem was not simply the absence of another model: the physical measurement intended to control amplitude was unstable under handling, and the production renderer did not acquire the ground deformation and object-grounding path needed to carry that mapping through. Interstitial relations were easier to retain because their consequence could remain a discrete object attached to a pair relationship and pass through the existing asset pipeline.

The two experiments therefore expose different limits of the same architecture. Extending tangible authorship beyond placement requires a physical property stable enough to carry the intended control and a digital representation capable of preserving its consequence. Where either side is missing, adding semantic or generative capability does not by itself make the physical action author the resulting space.

7 Ethical concerns

The project passed UAL's pre-ethics release and was classified as below minimal risk, so a full educational ethics application was not required. The approved plan included participant data collection, but the study was ultimately not conducted and no participant research data is reported here.

Privacy is partly determined by the system architecture. Speech recognition, text-to-speech and the conversational language model run locally. Object generation is the external-compute boundary: the shaped semantic prompt is sent to SDXL Turbo and SPAR3D running on a rented RunPod GPU. Voice recordings and the full conversation are not sent there. A deployed version handling sensitive descriptions would therefore need to treat object-generation prompts as data leaving the local machine rather than describe the system as private simply because most processing is local.

The generative models also make representational decisions that the person did not explicitly author. A semantic instruction can leave visual properties unspecified, and the generated asset may fill those gaps using patterns learned during model training. In Tangible Worlds this matters because the resulting object appears inside a scene whose semantics are otherwise presented as person-authored. The prompt-shaping layer was therefore constrained not to elaborate the subject, but the final visual form still contains model-authored choices. Ethical deployment would need to make that distinction clear rather than attribute every visible property of the generated world to the creator.

There is also an environmental cost. Development used both local inference and rented RTX 4090 compute, including experiments later discarded. I did not meter electricity use or retain enough GPU-runtime information to calculate a defensible carbon figure, so I do not estimate one retrospectively. Caching was introduced for reuse and latency, but it also avoids duplicate object generation. Future versions should measure compute use directly and assess whether each generative stage justifies its resource cost.

8 Conclusion and future work

This dissertation asked how tangible interfaces that translate physical arrangements into generative 3D environments might support new forms of spatial authorship and creative exploration. Tangible Worlds provides one answer through a deliberate separation of responsibilities. Semantically neutral physical props act as persistent handles for position and proximity; language assigns what those handles mean and controls other scene properties; generative models determine much of the resulting visual form. Spatial structure can therefore be rearranged without re-describing the scene, while meaning can be changed without replacing the physical prop.

Building the system showed that this separation cannot be treated as a simple division between human and AI control. Authorship has to be decided property by property. Automatic scale inference, speech recognition, prompt shaping and tracking each exposed a route by which an automatic process could silently acquire authority over a decision presented as belonging to the person**.** The resulting principle is that delegation itself is compatible with authorship when its boundary is explicit. Problems arise when that delegation is undeclared, including when sensing or interpretation errors effectively move an authored decision elsewhere.

The system also makes several dimensions independently revisable, but not equally so. Position, assigned meaning, scale and atmosphere can change without rebuilding the entire scene. Appearance is more constrained because a generated mesh is cached by meaning and the submitted interface has no ordinary reroll mechanism. Creative exploration is therefore an affordance of the architecture rather than an observed behavioural outcome**.** No creator study was completed, so this work does not establish whether people experience greater agency, ownership or exploratory freedom when using the system.

Future work should test those questions directly through qualitative creator sessions focused on how people interpret, revise and repair generated scenes. The interface should also support explicit regeneration of appearance while preserving meaning and spatial state. Beyond discrete objects, the terrain experiment suggests a route toward authoring continuous space once amplitude sensing and displaced-ground integration are made reliable. Interstitial relations remain a second route for extending spatial consequence, but their semantic and spatial quality across arbitrary object pairs still requires systematic evaluation.

9 LLM / Generative AI Disclosure

Generative AI tools were used during the development of this project as research and technical support tools. OpenAI ChatGPT and Anthropic Claude were used to research relevant neighbouring projects and technical concepts, investigate potentially suitable models and implementation approaches, explain unfamiliar programming and machine-learning concepts, and assist with debugging and code-related problem solving. This included support while exploring terrain-generation approaches and identifying candidate models for different parts of the system.

OpenAI Codex was used to assist with repeatable technical testing. During the speech-recognition evaluation, recorded audio samples were run through different model configurations and the resulting outputs were collected for comparison. During development of the atmosphere system, Codex was also used to execute combinations of test cases and capture screenshots of the resulting states for later comparison and evaluation.

AI-generated suggestions were treated as development assistance rather than as authoritative evidence. Technical decisions and project claims were checked against the implemented system, recorded outputs, project files and relevant source material.

10 Appendix A: Project artefacts and supplementary materials

The following materials accompany this dissertation and provide access to the implemented system, development evidence and project presentation.

Source code
GitHub repository: *https://git.arts.ac.uk/m-achakzai0320251/Tangible-Worlds-Thesis*

Development portfolio and evidence archive
Project portfolio: PORTFOLIO_1020

Project presentation
Presentation: *https://youtu.be/r9R8zmedLDQ*

Working system demonstration
Video demonstration: *https://youtu.be/XNu7JBYMaUo*

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